

Gastrointestinal physiology

Intestinal Phase & Accessory Digestive Organs

Integrated Biomedical and Clinical Sciences OMFS 512

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Learning Objectives

Learning Outcomes

1. Describe the components of the gastrointestinal system.
2. Understand the functional significance of the gastrointestinal system.
3. Compare and contrast major gastrointestinal secretions.

Outline

- Intestinal Phase
- Colonic Phase
- Accessory digestive organs

Gastrointestinal Tract

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graph TD; A[Gastrointestinal Tract] --> B[Primary Digestive Organs]; A --> C[Accessory digestive Organs]; B --> D["Oral cavity, esophagus, stomach, small and large intestine"]; C --> E["Salivary glands, liver, gall bladder and exocrine pancreas"]
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Primary Digestive Organs

“Oral cavity,
esophagus,
stomach,
small and large
intestine

Accessory digestive Organs

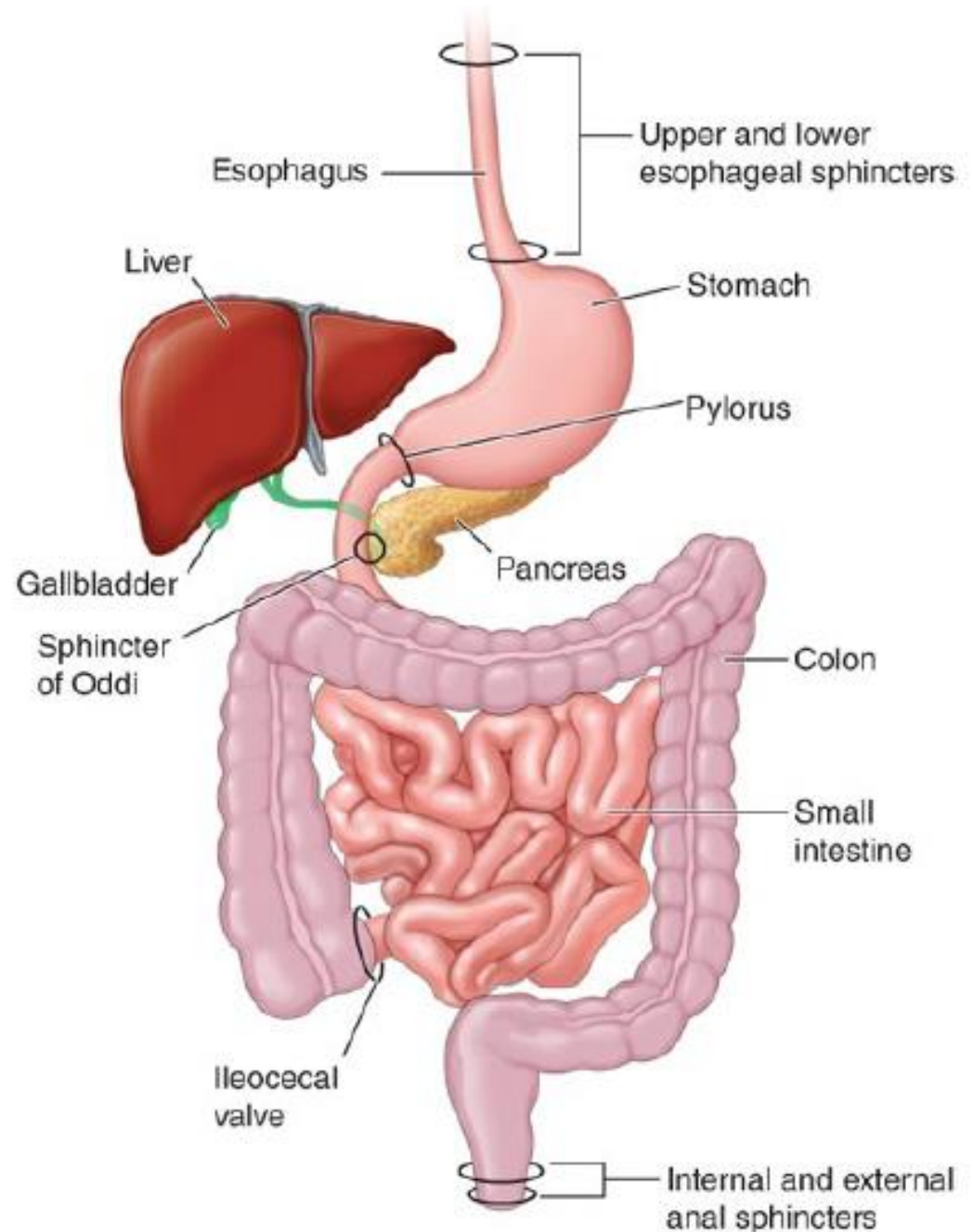
“Salivary glands,
liver, gall bladder
and exocrine
pancreas

Main Functions of GI System

- Ingestion
- Digestion
- Absorption
- Secretion

Phases Involved

- Cephalic phase
- Oral Phase
- Esophageal phase
- Gastric phase
- Intestinal and colonic phase

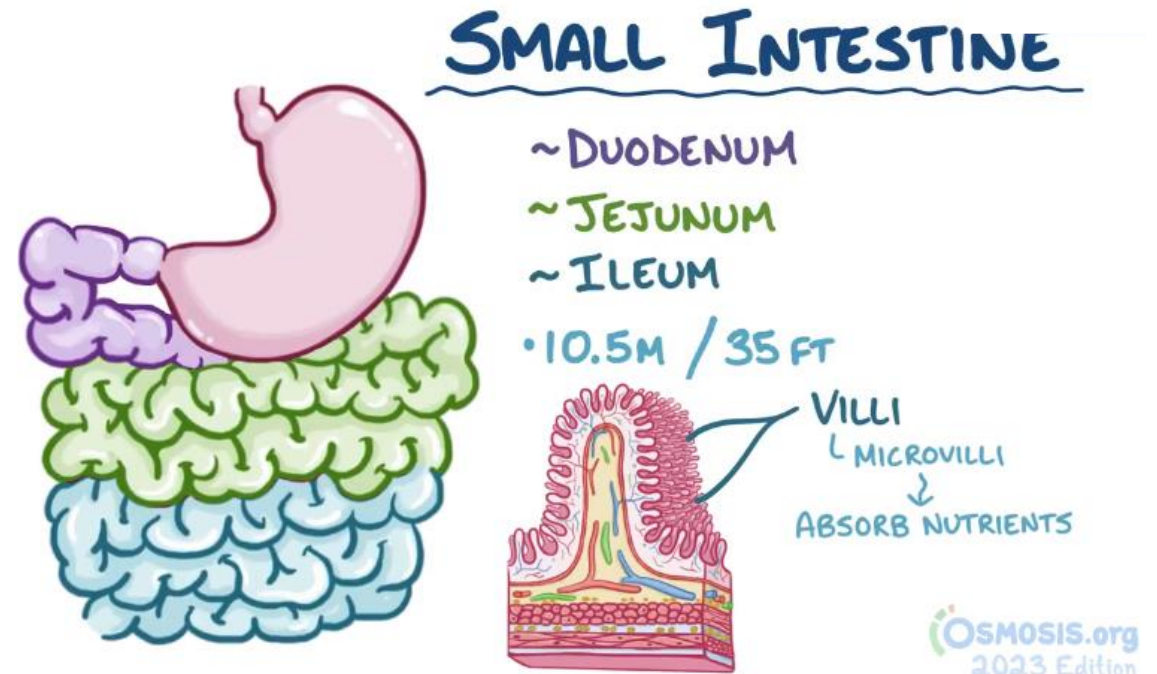




The Intestinal Phase

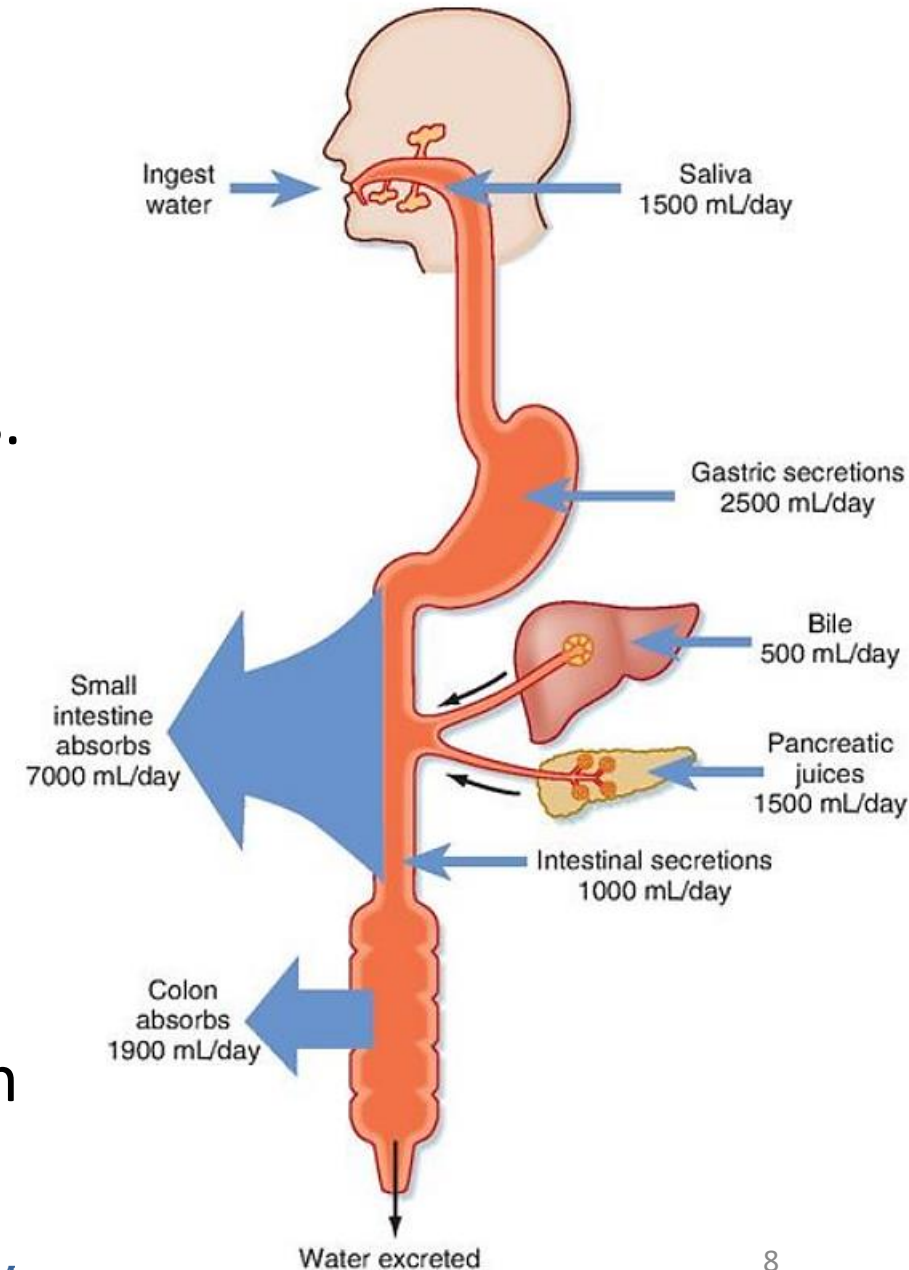
Small Intestines

- Longest portion of GI tract
- Location
 - Inferior portion of abdominal cavity, caudal to stomach
 - Between gastric pylorus and ileocecal junction
- Function
 - Main site for food digestion and nutrient absorption
 - Extracts excess water
 - Sends food waste to large intestine
- Main Segments
 - Duodenum, Jejunum, Ileum

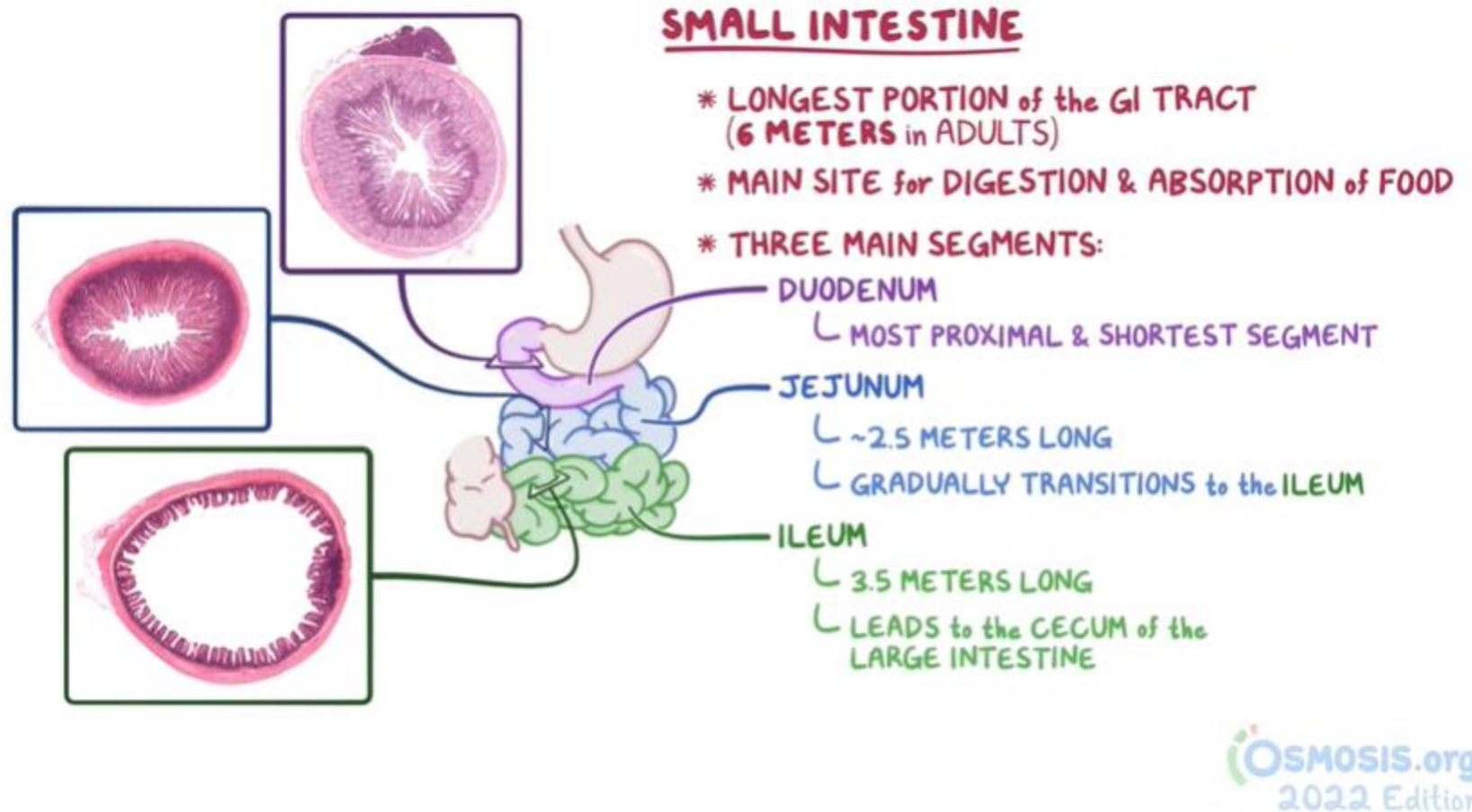


Function of the Small Intestine

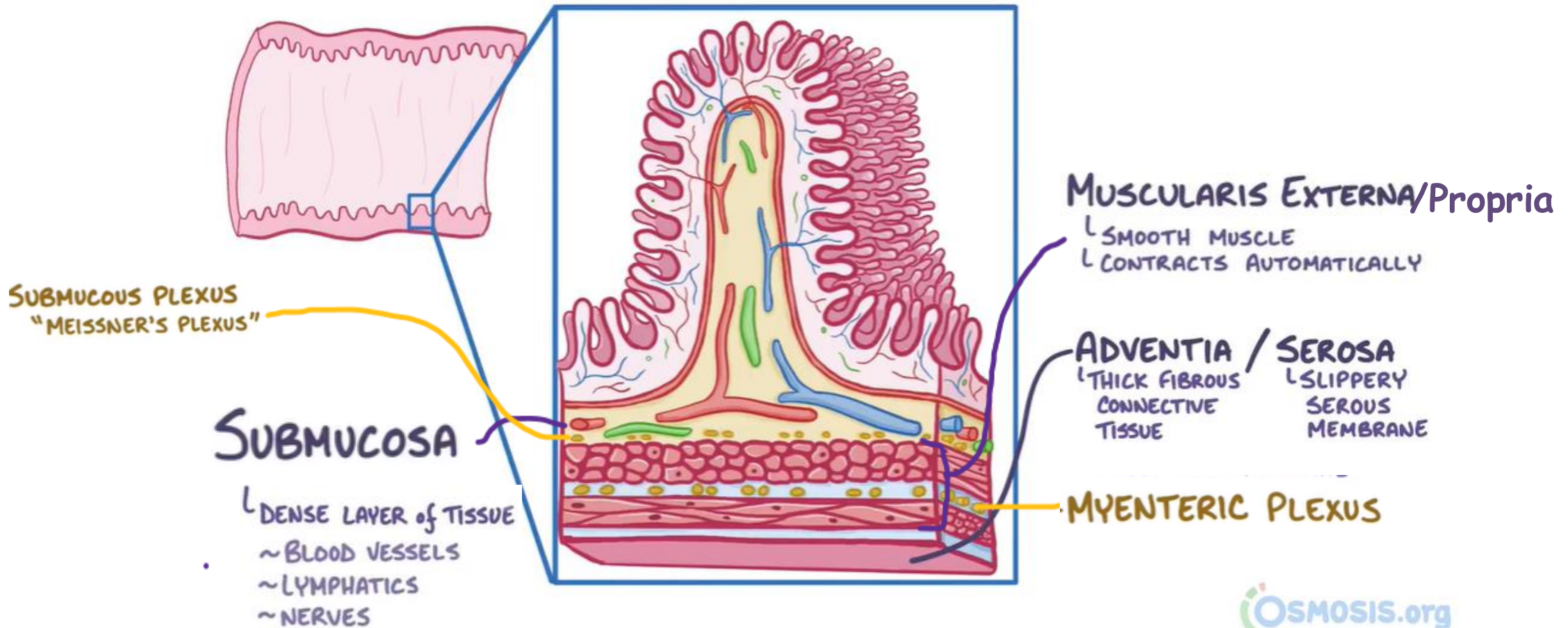
- 1- Secretion of intestinal juice.
- 2- Chemical digestion of carbohydrates, protein and fats with the aid of pancreatic secretions and bile acids.
- 3- Absorption of nutrients and water. The daily fluid load is around 9 L (around 7 L is endogenous GI secretions). Most of this fluid (6-7 L) is absorbed back by the small intestine (with nutrients)
- 4- Secretion of the hormones like cholecystokinin and secretin.
- 5- Immune-protective function via the action of Paneth cells and Peyer's patches.



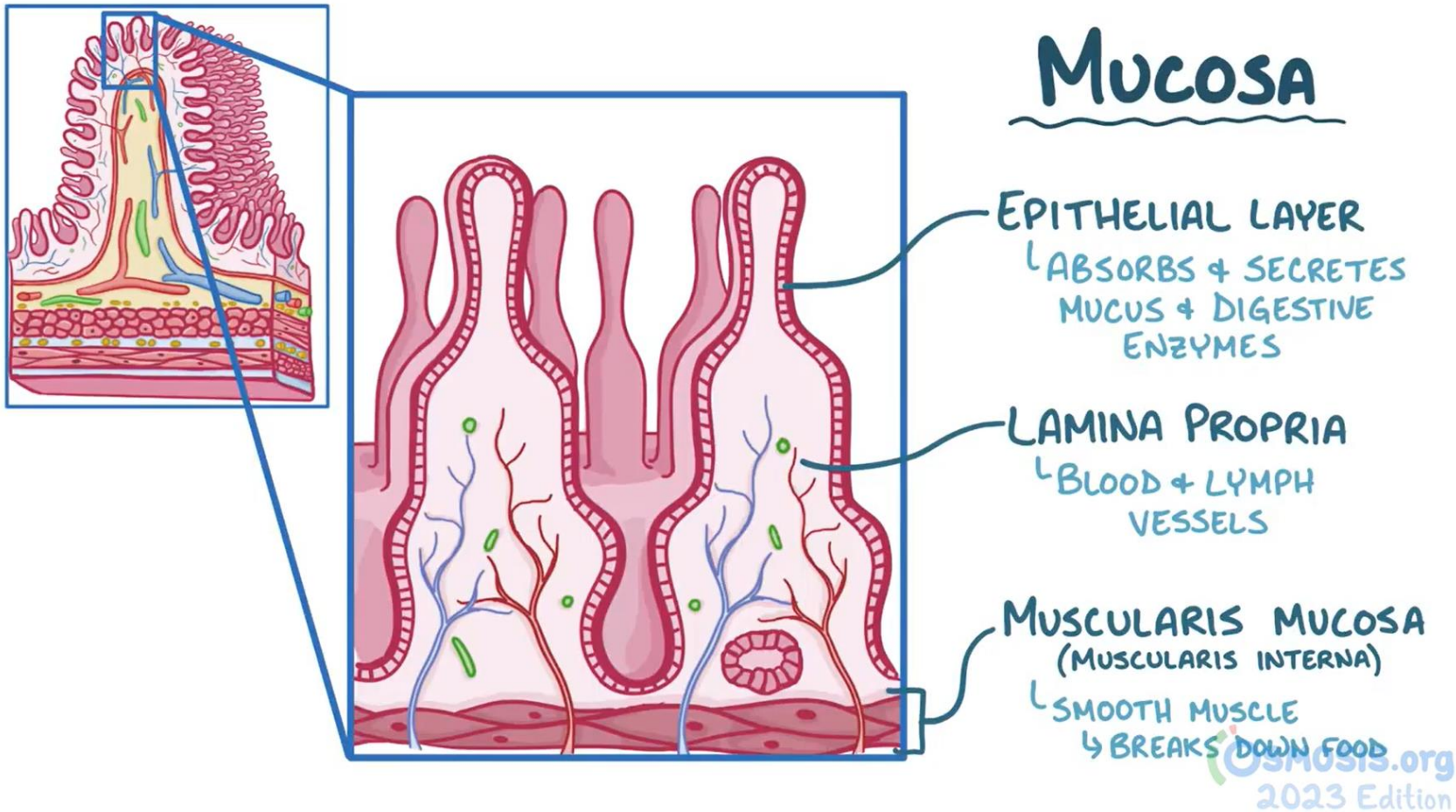
Small Intestine Anatomy

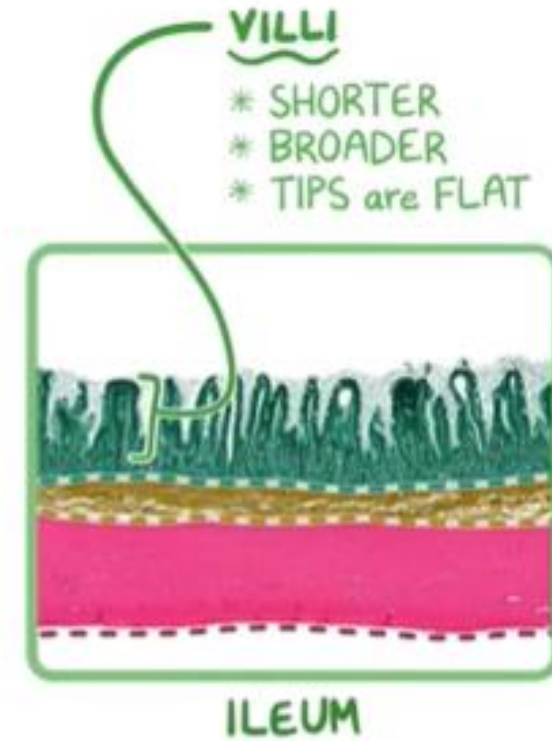
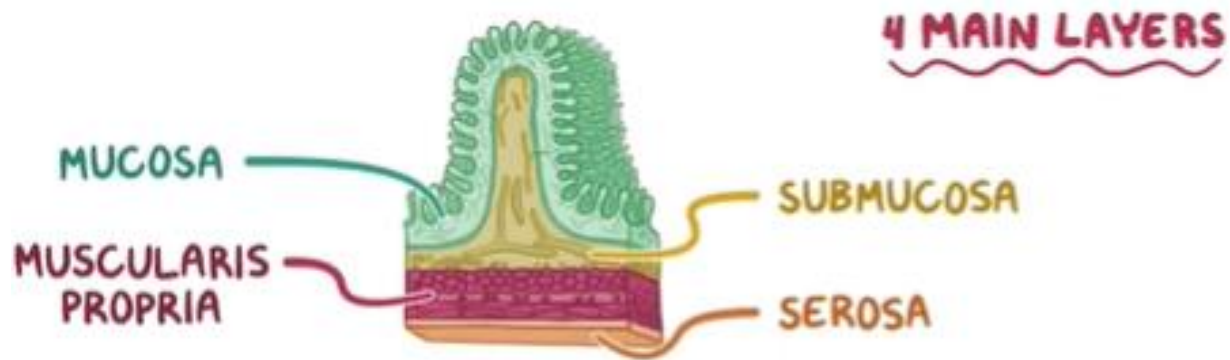


GI Tract- 4 Main Cellular Layers

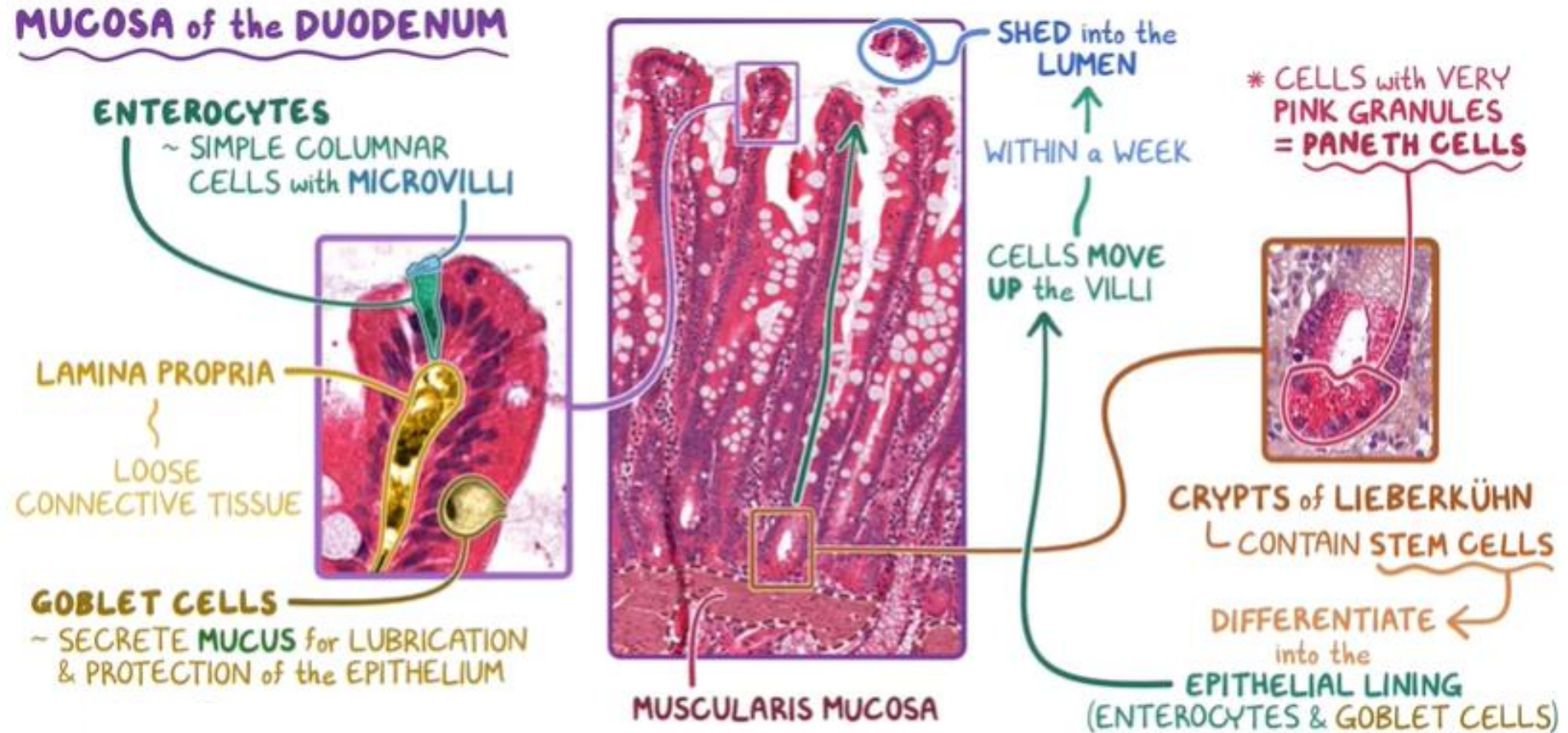


GI Tract Layers (Cont.)





Mucosa of the Duodenum



Mucosa of the Ileum

Cell Mol Gastroenterol Hepatol. 2018 Dec 14;7(3):503–513. doi: [10.1016/j.jcmgh.2018.12.002](https://doi.org/10.1016/j.jcmgh.2018.12.002)

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Figure 1.



SHORTER,
BROADER VILLI



LACTEALS

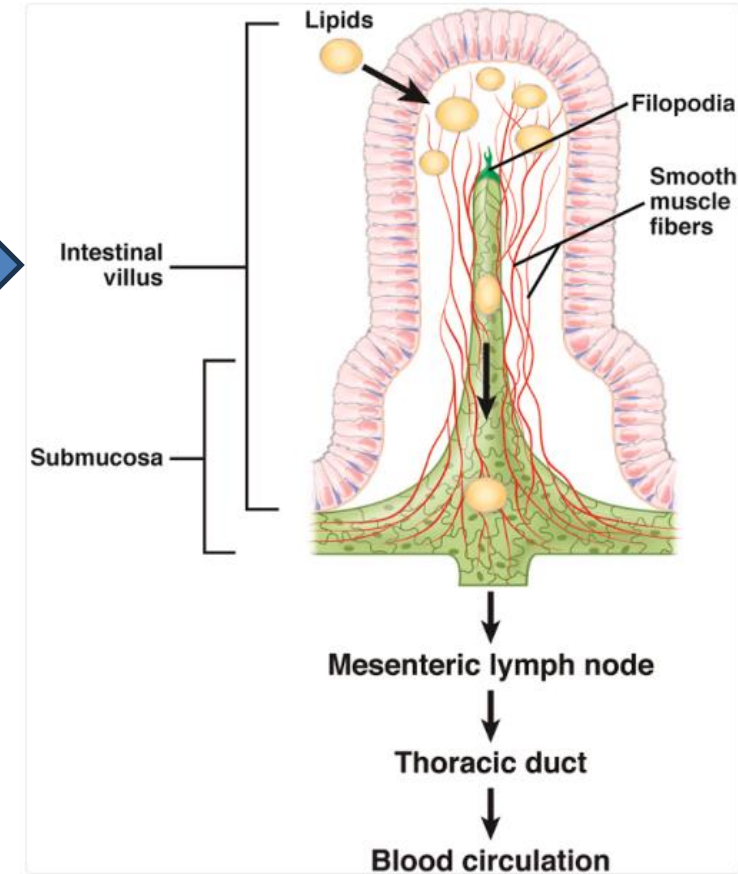
- ~ WHITE SPACES within the CORES of some VILLI
- ~ LYMPHATIC CAPILLARIES
- ↳ ABSORB DIETARY FATS



PEYER'S PATCHES

- ~ MASSES of LYMPHOID TISSUE within the LAMINA PROPRIA

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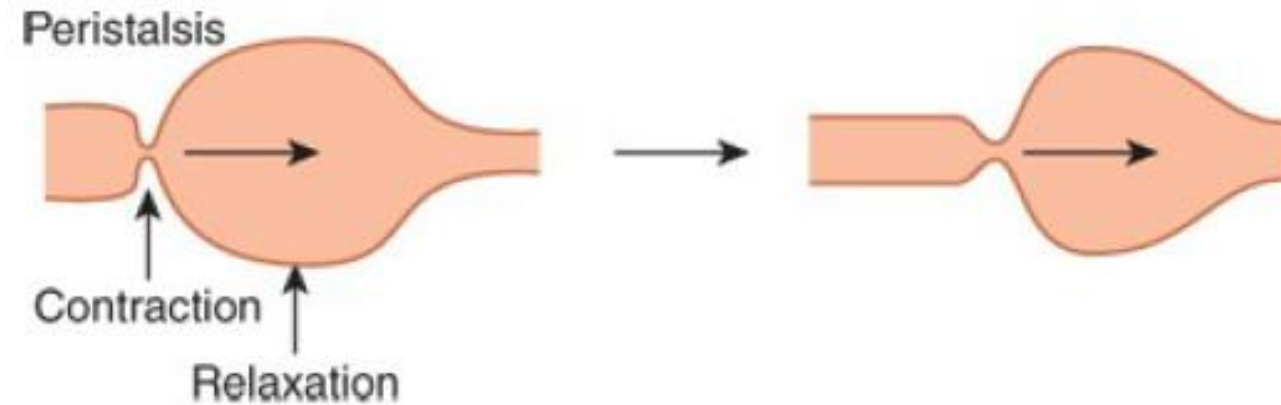


Defining Characteristics

- •The proximal portion of the duodenum is the most common site of duodenal ulcers due to its proximity to gastric acid.
- •Duodenum has **Brunner's glands in the submucosa**. Brunner's glands are branched glands that produce alkaline mucous (pH 8-9). It aims to neutralize the acidic chyme that enters from the stomach.
. .
- •**The jejunum** is characterized by numerous, long villi. The jejunum is the major area for final digestion and absorption of nutrients in the small intestine.
- •**The ileum** is the longest portion of the small intestine. It is characterized by Peyer's patches and short villi. Peyer's patches are lymphoid follicles that play an important role in immune surveillance (sampling foreign materials) and immune response

Intestinal Motility- Peristalsis

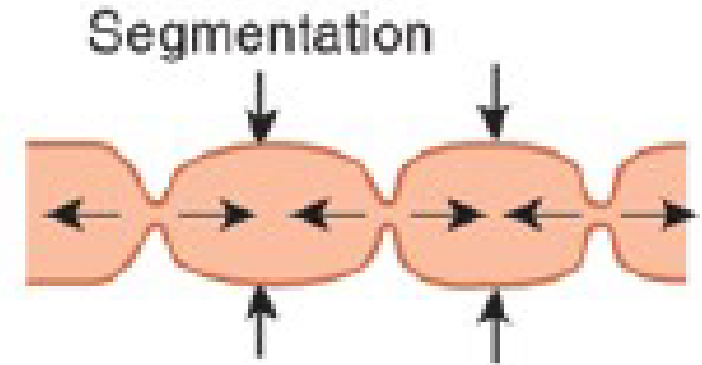
- Peristaltic contraction
 - Reflex response initiated by gut wall stretching
 - Circular contraction behind the chyme prevents backflow
 - Longitudinal relaxation in front of the chyme propels it forward through the tract
 - Controlled by autonomic input and is independent of extrinsic innervation



Source: Kim E. Barrett, Susan M. Barman, Heddwen L. Brooks, Jason X.-J. Yuan: Ganong's Review of Medical Physiology, 27th Edition
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Intestinal Motility- Segmentation

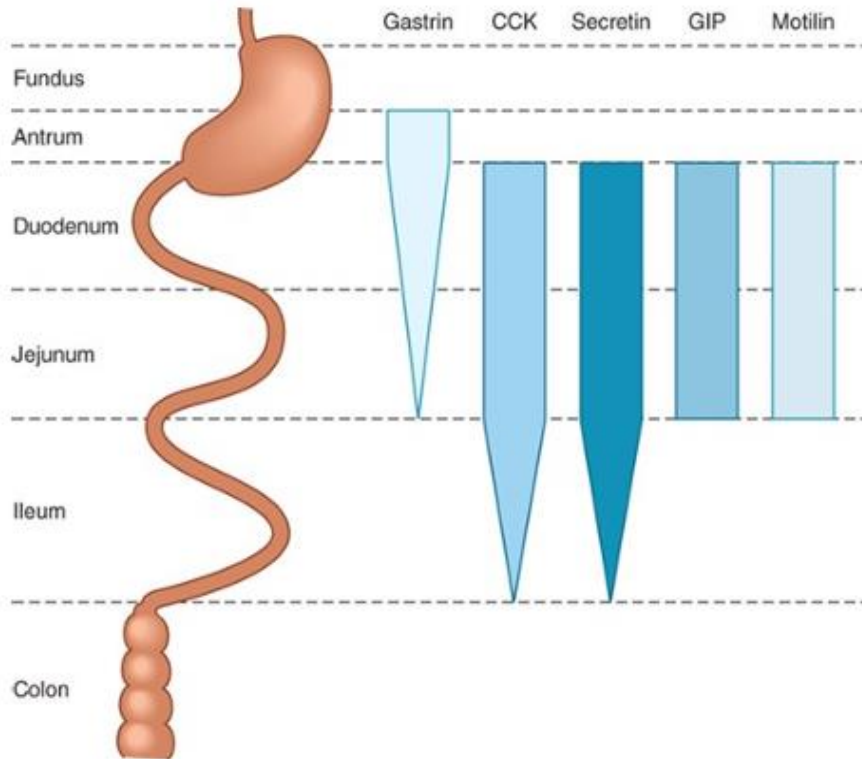
- Segmentation contraction
 - Functions to mix the contents of the lumen
 - Occurs in a short span of the tract
 - Contraction moves the chyme back and forth continually until nutrients have been absorbed



Intestinal Secretions- Enteroendocrine Cells

- Enteroendocrine cells are hormone secreting cells found in the mucosa of the stomach, small intestine, and colon.
 - Some secrete a single peptide and are named after a letter (S cells, G cells, etc.)
 - Some are producers of serotonin (enterochromaffin cells)
 - Some are producers of histamine (ECL cells)

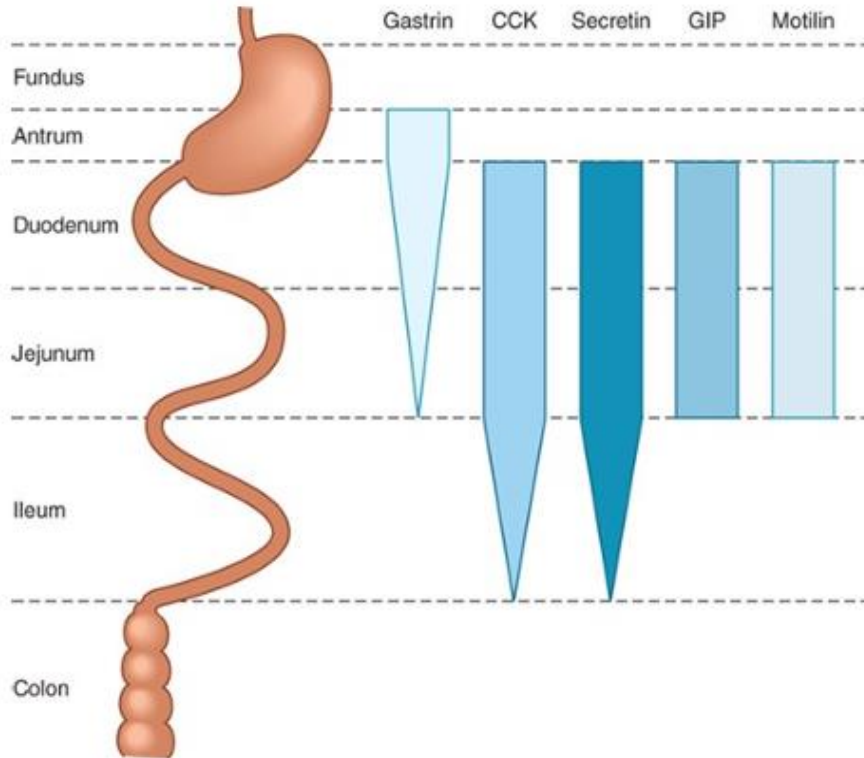
Intestinal Secretions- GI Hormones



Source: Kim E. Barrett, Susan M. Barman, Heddwen L. Brooks, Jason X.-J. Yuan: Ganong's Review of Medical Physiology, 27th Edition Copyright © McGraw Hill. All rights reserved.

- **Gastrin** is produced by G cells.
 - Stimulates gastric acid and pepsin secretion and stimulate growth of the mucosa of the stomach and small and large intestines
- **Cholecystokinin (CCK)** is produced by I cells.
 - Stimulates secretion of pancreatic enzymes, stimulate gall bladder contraction, and relax the sphincter of Oddi- allowing bile and pancreatic juices to flow into the intestinal lumen, among other functions

Intestinal Secretions- GI Hormones



Source: Kim E. Barrett, Susan M. Barman, Heddwen L. Brooks, Jason X.-J. Yuan: Ganong's Review of Medical Physiology, 27th Edition Copyright © McGraw Hill. All rights reserved.

- **Secretin** is produced by S cells.
 - Increases the secretion of bicarbonate by the pancreatic duct cells and the biliary tract
- **Gastric Inhibitor Peptide (GIP)** is produced by K cells.
 - In large amounts it inhibits gastric secretion and motility and at physiological levels (levels seen after a meal) it stimulates insulin secretion
- **Motilin** is secreted by enterochromaffin cells.
 - Produces smooth muscle contraction in the period between meals

Intestinal Secretions

- Brunner's glands secrete **alkaline mucus**.
 - This mucus contains excess bicarbonate to help in neutralizing the hydrochloric acid entering the duodenum from the stomach.
- The intestinal epithelia (enterocytes) produce disaccharidases.
 - sucrase, maltase, and lactase
 - Function to break down disaccharides into monosaccharides.

Other secretions that aid in the regulation of GI function including

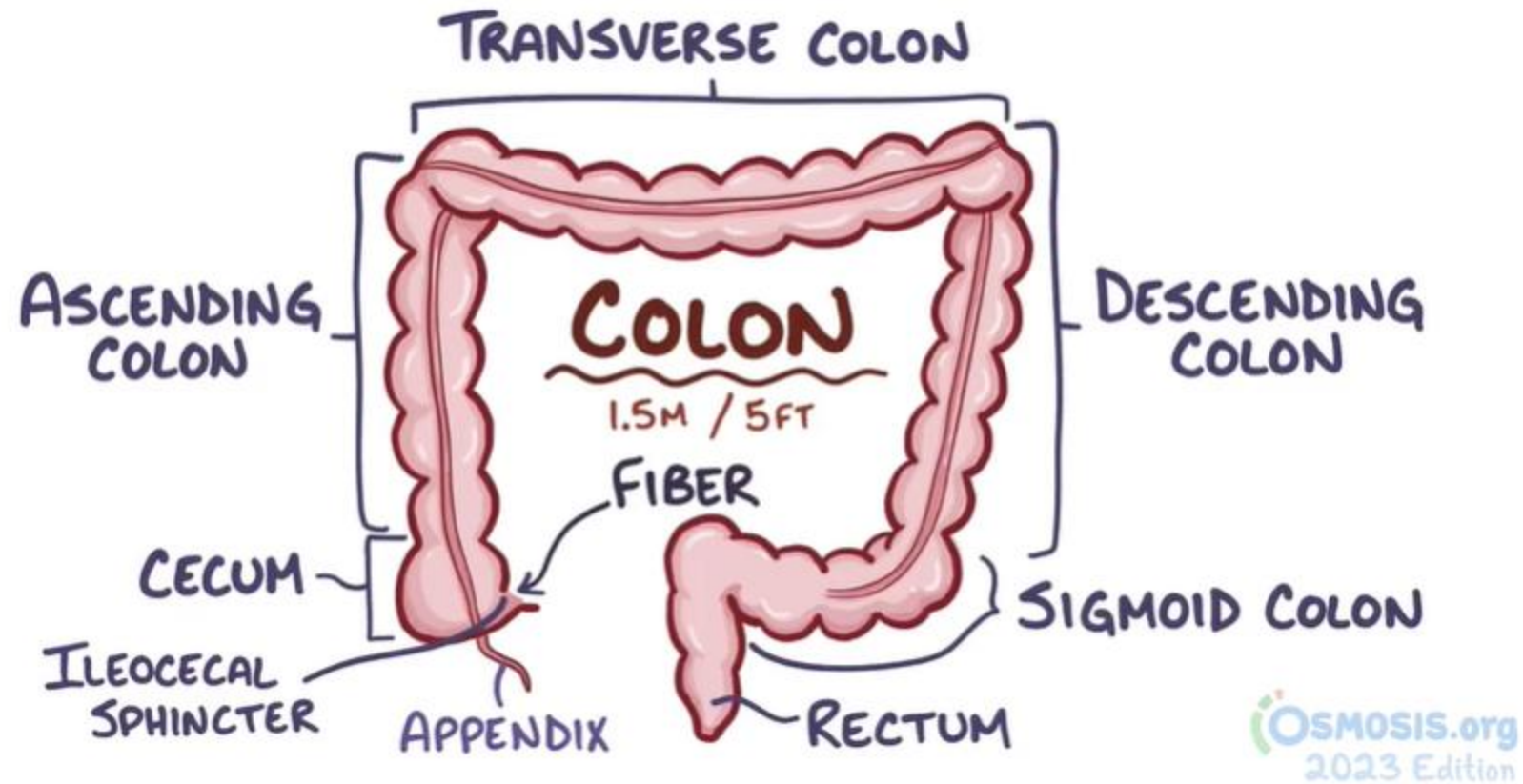
- –Enteroglucagon (aka, glucagon-like peptide 2 from L cells, **trophic enzyme**).
- –Enteropeptidase (aka, enterokinase), secreted in the duodenum to convert **trypsinogen to trypsin**.
- Glycocalyx is a glycoprotein coat that covers the microvilli below the mucus layer. Glycocalyx contains hydrolytic enzymes like enteropeptidases, dipeptidases, and disaccharidases.

Intestinal Absorption

- The small intestine is the main site of nutrient and water absorption.
- Most absorption occur in the jejunum or along the length of the small intestine. Exceptions include:
 - Absorption of vitamin B12-intrinsic factor complex and bile acids takes place in the ileum.
 - Iron is absorbed in the duodenum.

The Colonic Phase

Large Intestine Anatomy

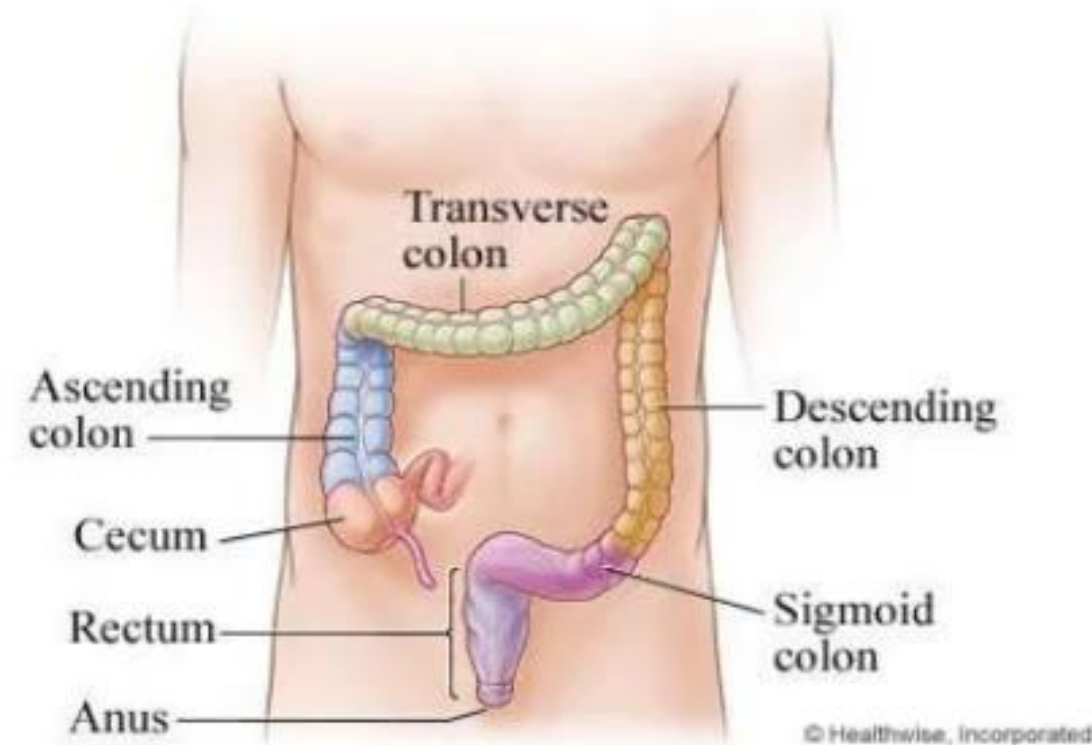


Functions of the Large intestine

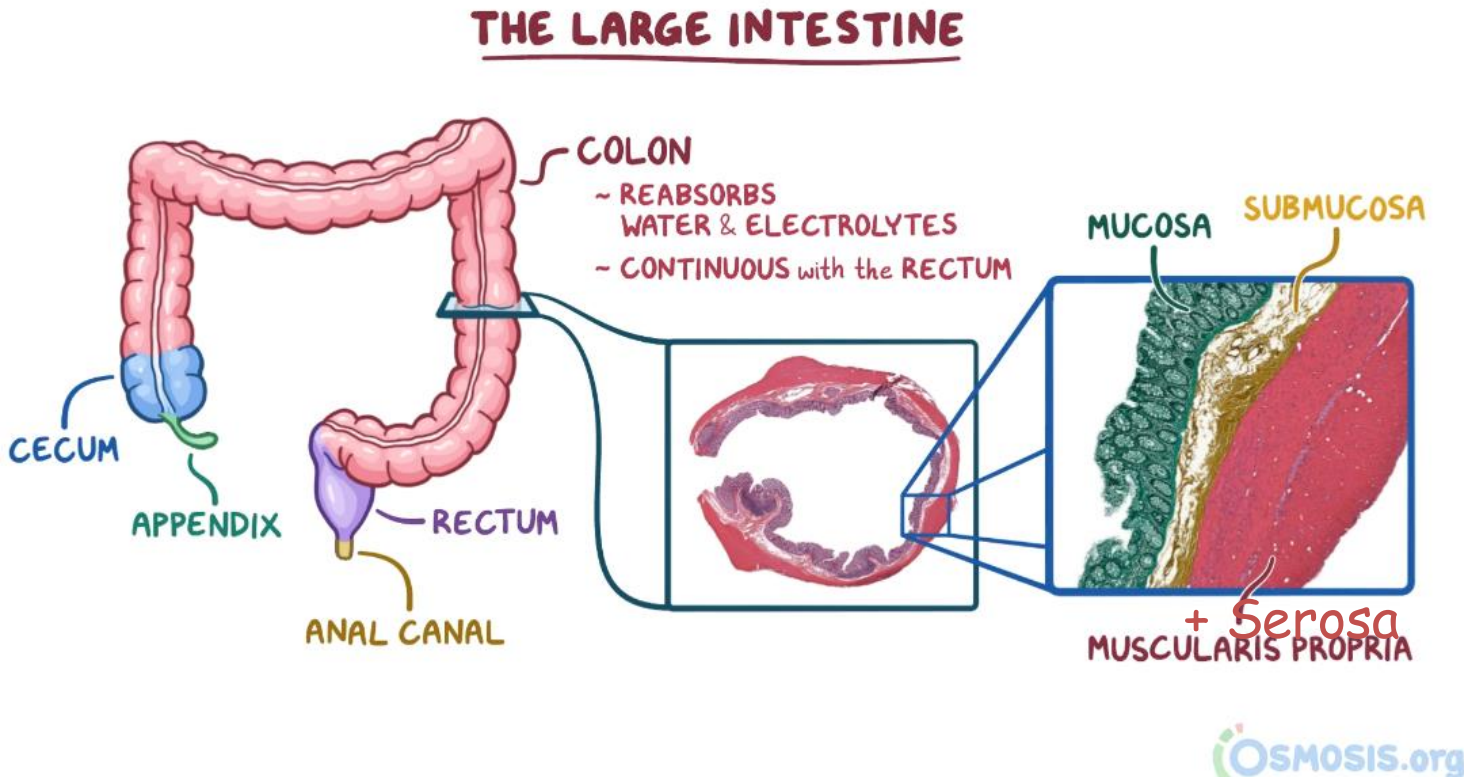
- Complete water and electrolyte absorption.
 - The ascending colon absorbs the remaining water and other key nutrients and form the indigestible material mass.
 - Sodium ions and water molecules are transported from the intestinal lumen into the blood.
- Production and absorption of vitamins by the gut microbiome.
 - Vitamin K and B are produced by the colonic bacteria and can be absorbed into the blood from the colon.
- Elimination

- Ascending colon
 - Absorb remaining water and other key nutrients from indigestible material
 - Solidify indigestible material to form stool
- Descending Colon
 - Stores feces until they are emptied into the rectum
- Sigmoid Colon
 - Contracts to increase pressure inside the colon and push the stool toward the rectum
- Rectum
 - Holds feces until eliminated via defecation

Large Intestine



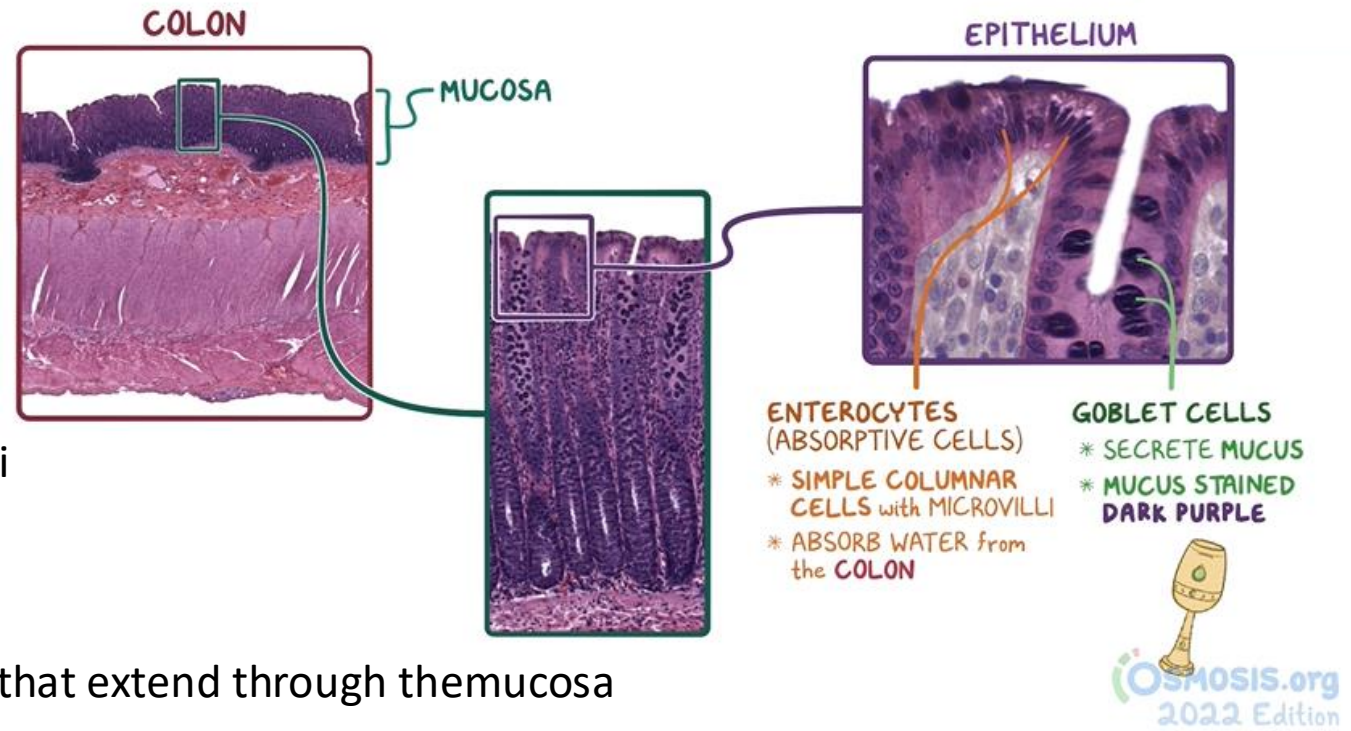
Colon Histology



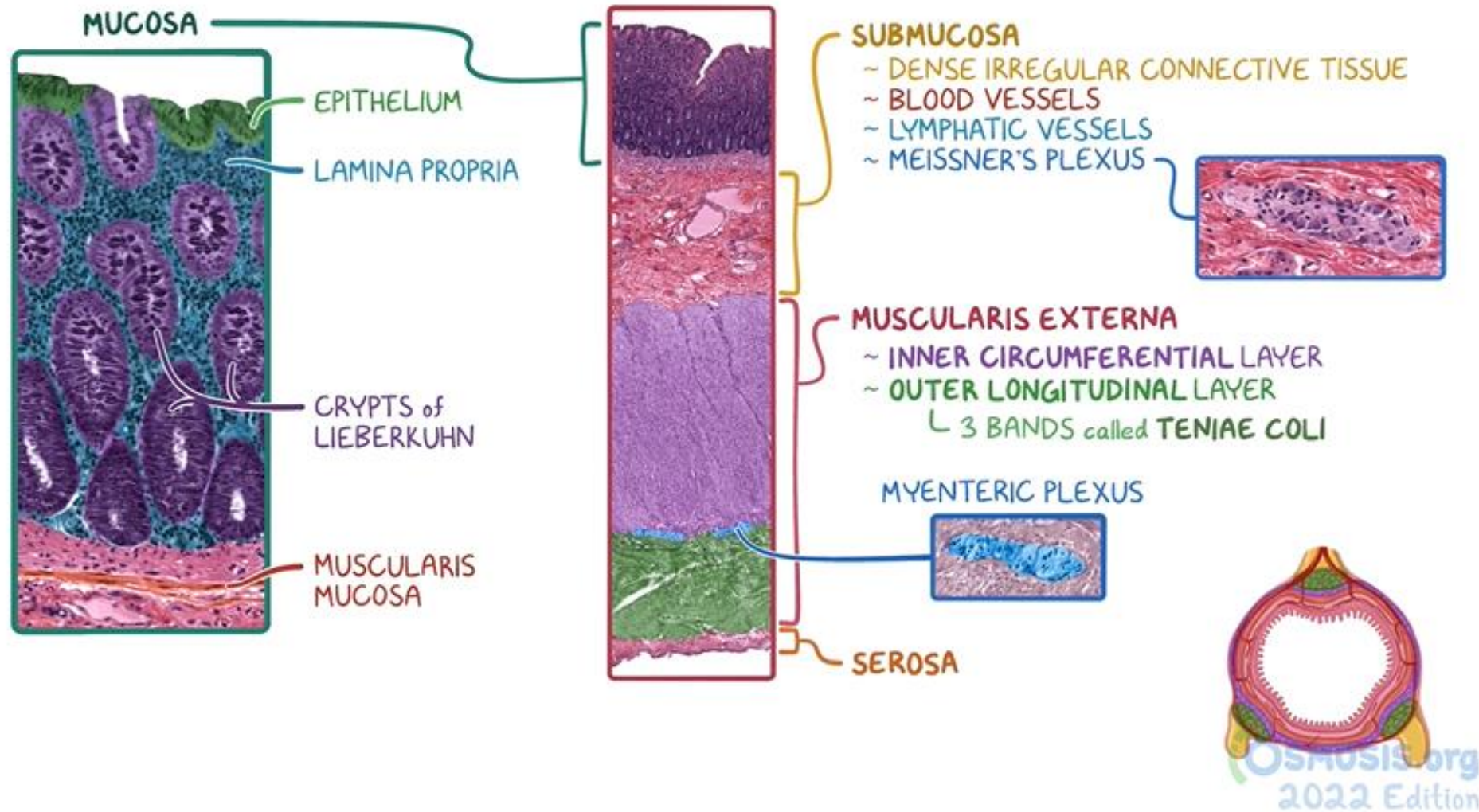
- Contains same 4 layers
 - Mucosa
 - Submucosa
 - Muscularis propria (externa)
 - Serosa

Colon Histology- Mucosa

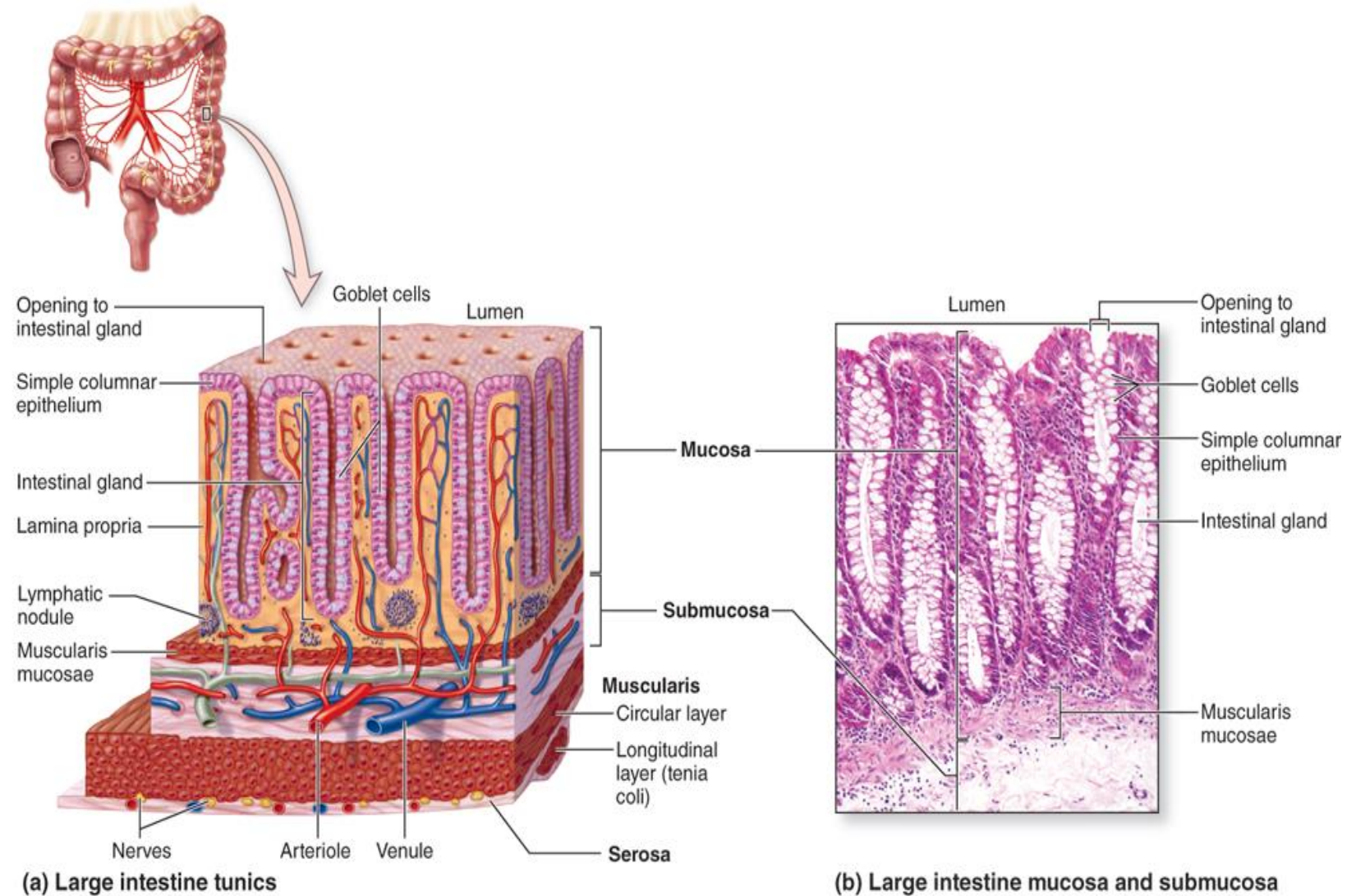
- No long villi
- 1st layer of cells = epithelium
 - **Goblet cells**
 - Secrete mucus
 - **Enterocytes** (Absorptive cells)
 - Simple columnar cells with microvilli
 - Absorb water from colon
 - **Crypts of Lieberkhun**
 - Continuous, straight tubular glands that extend through the mucosa
 - Higher concentration of goblet cells superficially and enterocytes in deeper portions of crypts
- 2nd layer = Lamina Propria
 - Tissue between crypts and epithelium
 - Contains immune cells
- Deepest layer = Muscularis Mucosa



Colon Histology



- (An additional diagram to aid in visualization of the large intestine cellular structure)



Source: Anthony L. Mescher:
Junqueira's Basic Histology: Text and Atlas, 17th Edition
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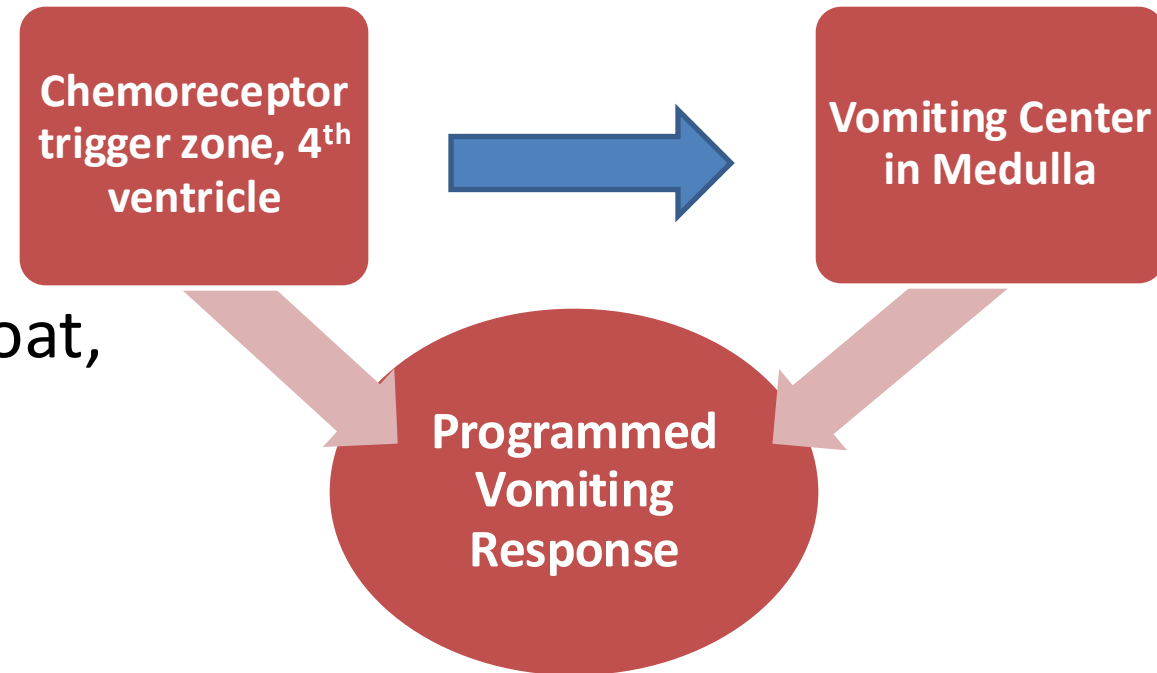
Intestinal Pathophysiology

Vomiting Reflex

Vomiting is controlled by the vomiting center of the medulla.

Etiology

- The vomiting center in the medulla is stimulated by tickling the back of the throat, gastric distention (overeating), and vestibular stimulation (motion sickness).
- The chemoreceptor trigger zone in the fourth ventricle is activated by GI irritation (medication/emetics, toxins, pregnancy,..), radiation & motion sickness.



Mechanism:

A wave of reverse peristalsis begins from the small intestine and move the GI contents towards the stomach and esophagus. If the pressure in the esophagus becomes high enough to open the upper esophageal sphincter, vomiting occurs.

Complications:

Dehydration, hypokalemia, metabolic alkalosis, aspiration.

Malabsorption Syndromes

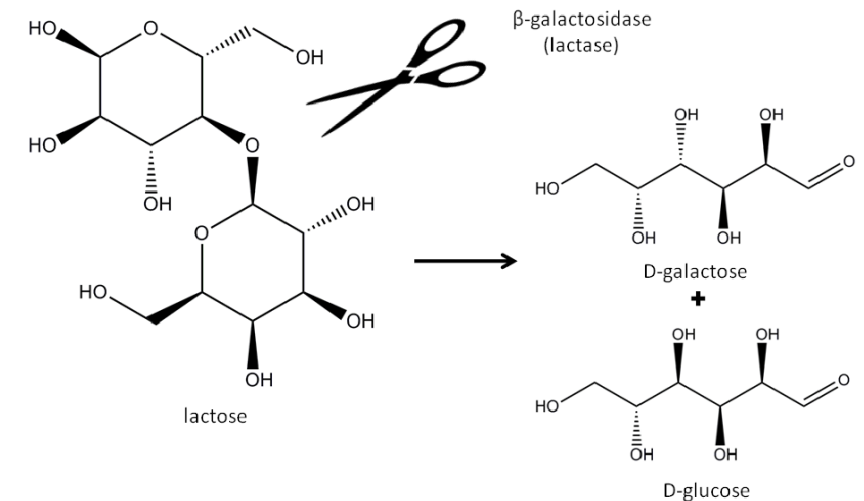
Lactose intolerance

Lactose is digested in the intestine by lactase enzyme to yield glucose and galactose. Deficiency of lactase enzyme can lead to lactose intolerance.

Etiology: deficiency of lactase enzyme.

Symptoms: cramps, diarrhea, flatulence and bloating after ingesting dairy products.

Biochemical mechanism: undigested lactose undergo bacterial fermentation. This process generates carbon dioxide and hydrogen molecules resulting in flatulence. Undigested products create an osmotic gradient leading to diarrhea.



Lactose Intolerance- Classification

1. **Congenital alactasia:** diagnosed soon after birth. There is a complete absence of lactase enzyme due to autosomal recessive disorder. Both parents should be carriers for the defective gene.
2. **Acquired:**
 1. Primary lactose intolerance. Develops over time. Often observed in adulthood when lactase production decreases.
 2. Secondary lactose intolerance. Develops secondary to disease or infections. For example, injury like celiac disease, or in cases of chemotherapy treatment. Secondary to gastroenteritis (more common with rotavirus infection).

Lactose Intolerance- Risk factors:

- 1- Advanced age.
- 2- Ethnicity: It is more common in individuals of African, Asian, Hispanic and American Indian descent.
- 3- Premature birth: reduced levels of lactase because lactase-producing cells don't fully develop until late phase of pregnancy.
- 4- Diseases affecting the small intestine including bacterial overgrowth, celiac disease and Crohn's disease.
- 5- Cancer treatments including irradiation therapy for stomach cancer or treatment with chemotherapeutic agents can increase the risk of developing lactose intolerance.

Management: Avoidance of lactose containing products. OTC lactase tablets (Lactaid®).

Steatorrhea

- Steatorrhea is a form of fat malabsorption characterized by pale, foul-smelling oily stools.
- Symptoms:
 - Chronic diarrhea, abdominal discomfort and bloating.
- Complications
 - Deficiencies of fat-soluble vitamins (A, D, E, and K).
 - Weight loss and consequences of malnutrition.
 - Liver cirrhosis, seizures.

Steatorrhea- Etiology:

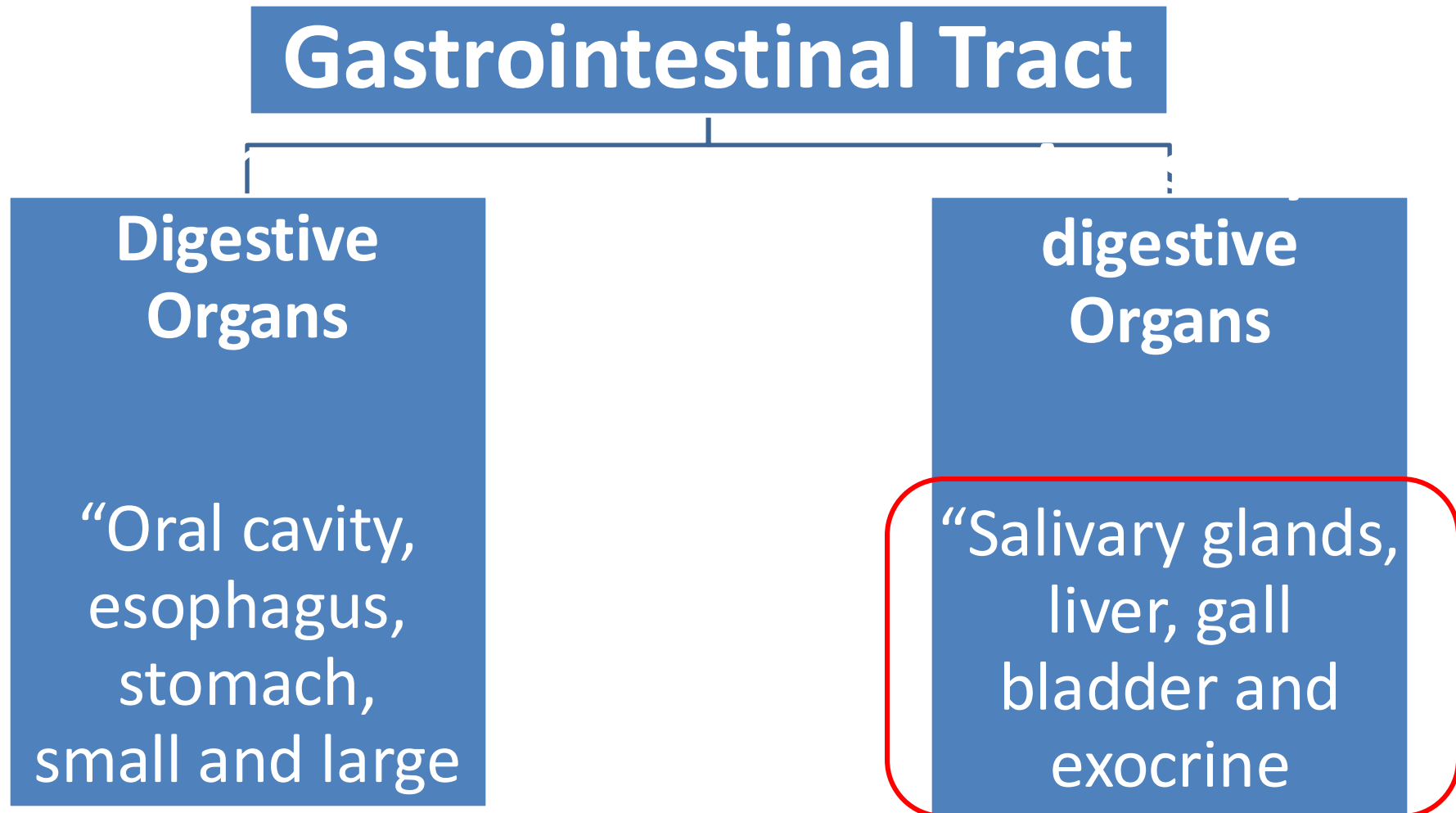
1. Exocrine pancreatic insufficiency (EPI) leading to impaired digestion of fat due to lipase deficiency. Can occur secondary to cystic fibrosis.
2. Bile acid deficiency. The inability to absorb bile acids in the distal ileum will result in diminished bile acid pool.
3. Genetic deficiency in micelle forming proteins.
4. Diseases affecting the small intestine, like *SIBO*.

SIBO “small intestinal bacterial overgrowth”

Overgrowth of bacterial population in the small intestine.

It slows down the passage of food and waste products in the digestive tract leading to loss of appetite, abdominal pain, bloating, diarrhea.

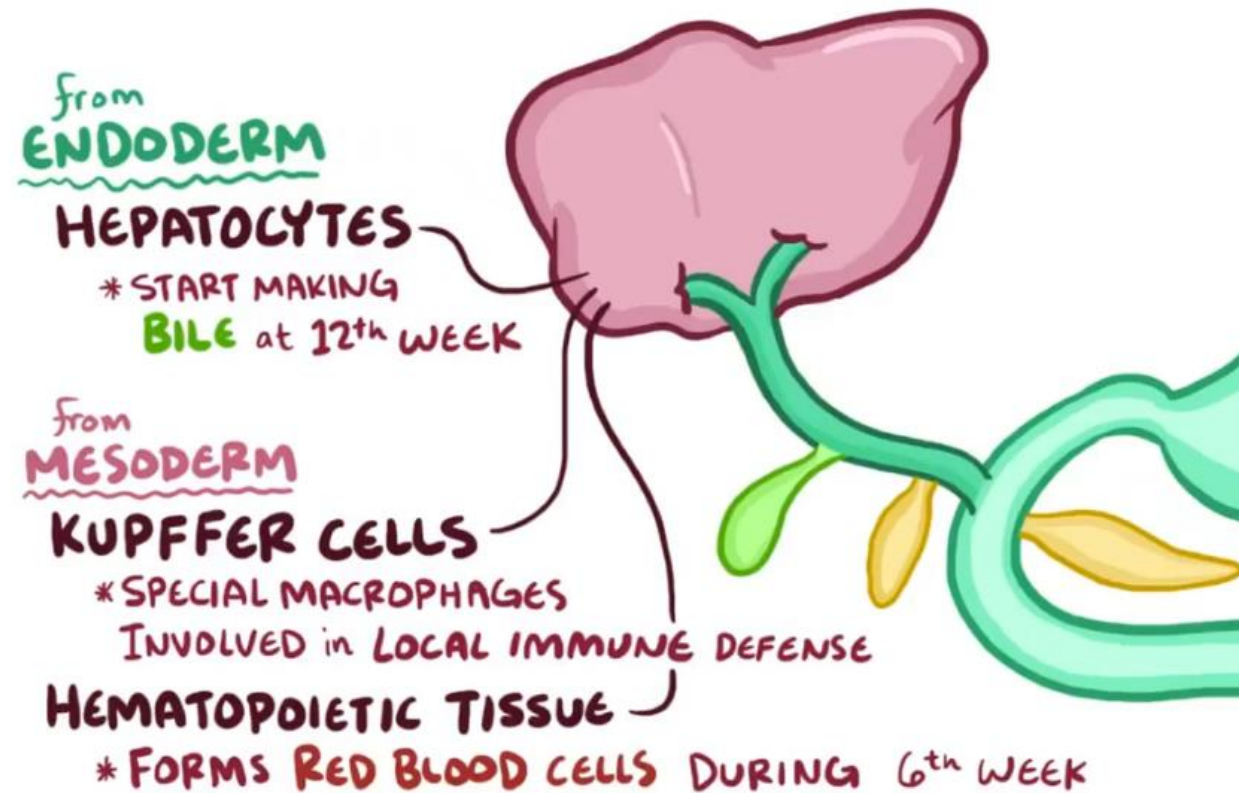
Accessory Digestive Organs



Liver and Gall Bladder

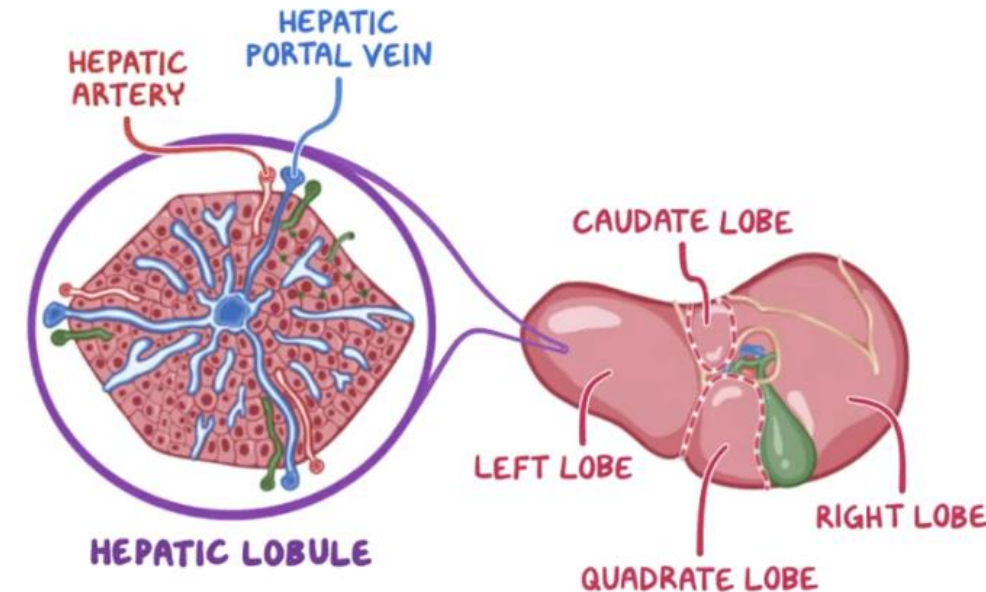
Embryonic development

- The liver, gallbladder, and biliary system arise from the liver bud (foregut) at W4.
- Endoderm develop into the hepatocytes, mesoderm yield Kupffer cells, connective and hematopoietic tissues.



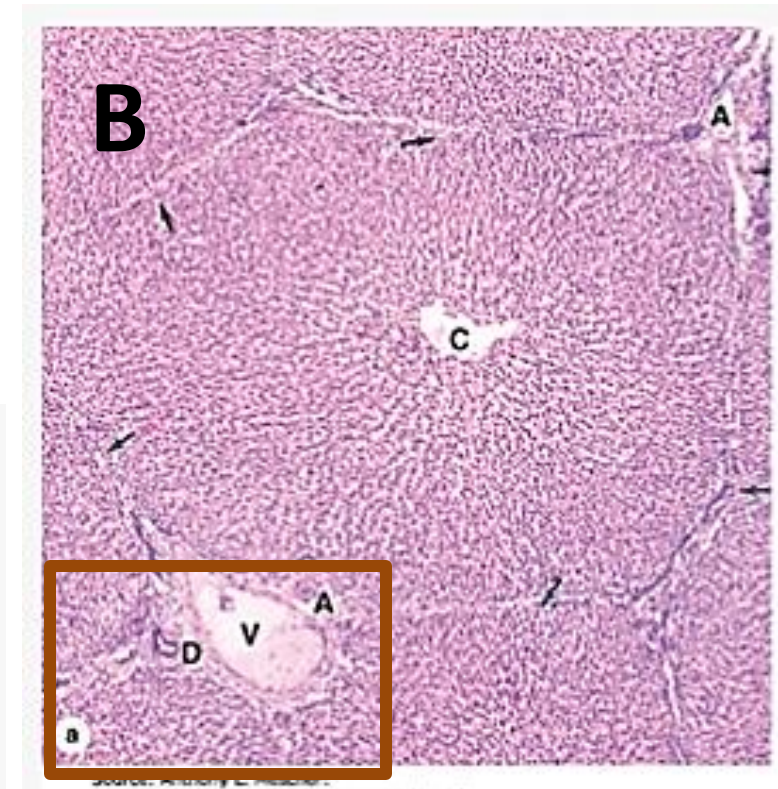
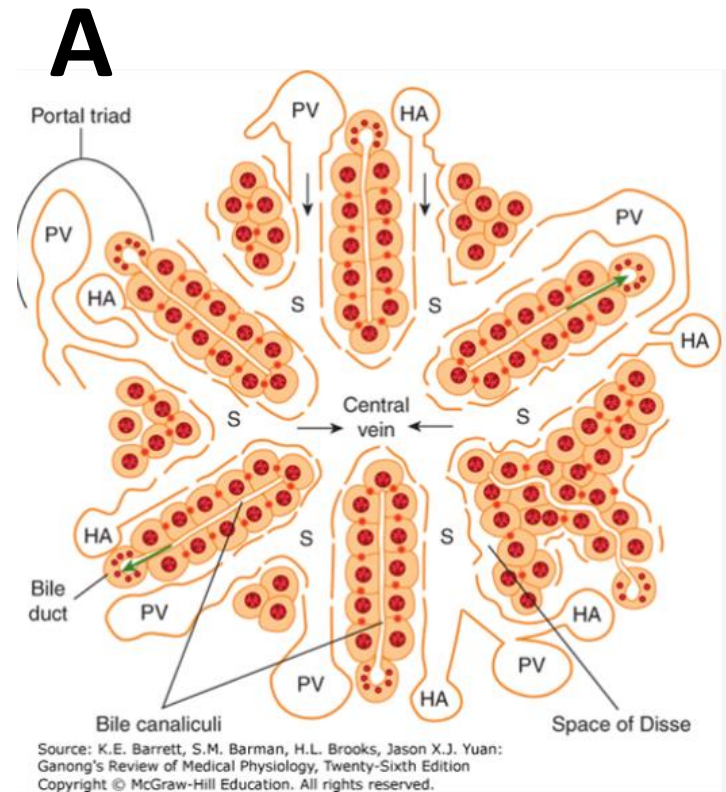
Liver

- Largest internal organ, in adults averaging about 2% of the body.
- Liver is enclosed by a thin layer of fibrous connective tissue “ Glisson’s capsule”.
- The liver is composed of four lobes, incompletely separated by deep fissure.
- The functional unit of the liver is the lobule.



Lobule Histology

In the liver lobules, hepatocytes are arranged in columns radiating from a central vein. Portal triad is found at the corner of each lobule.

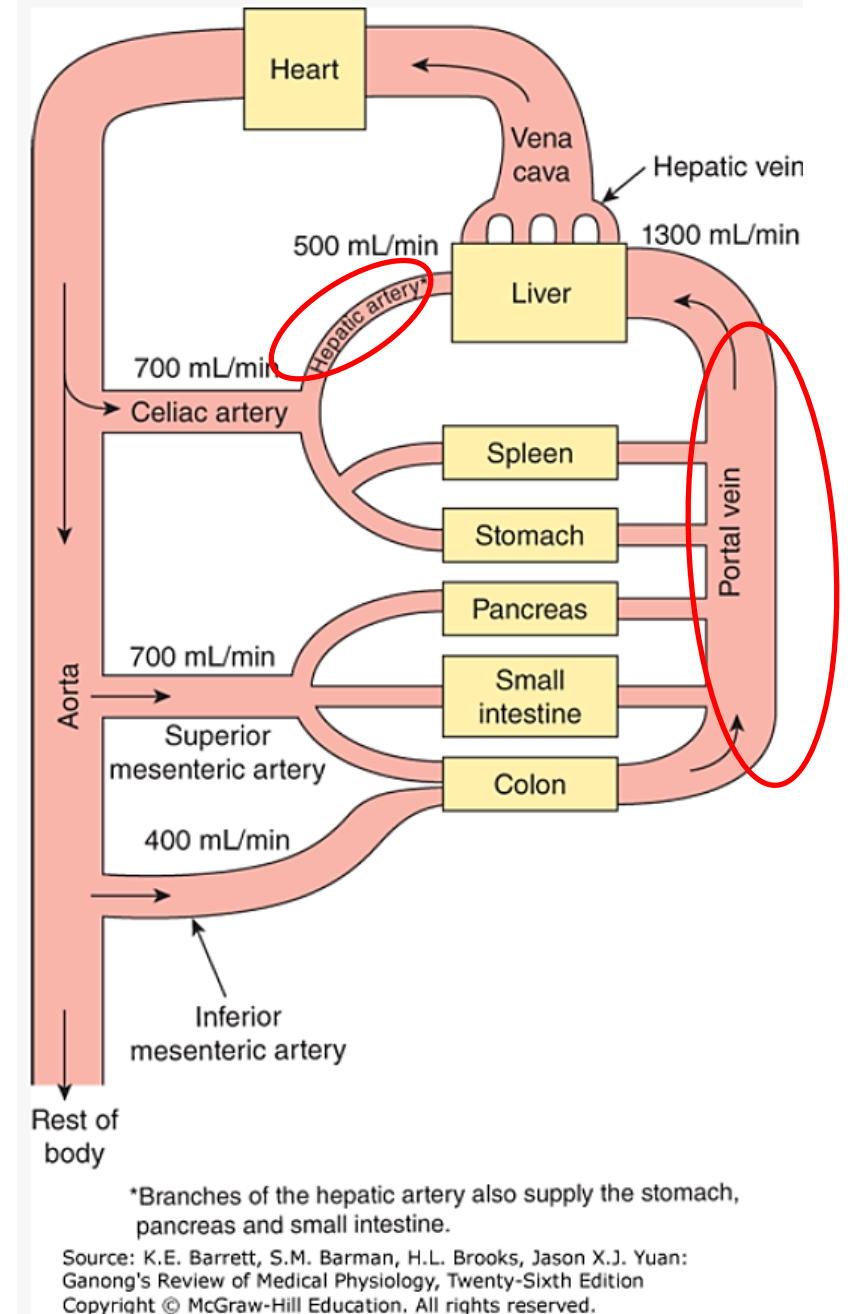


Portal Triad (venule, arteriole and bile ductule)

Blood supply

The liver receives dual blood supply, 25 % of the cardiac output:

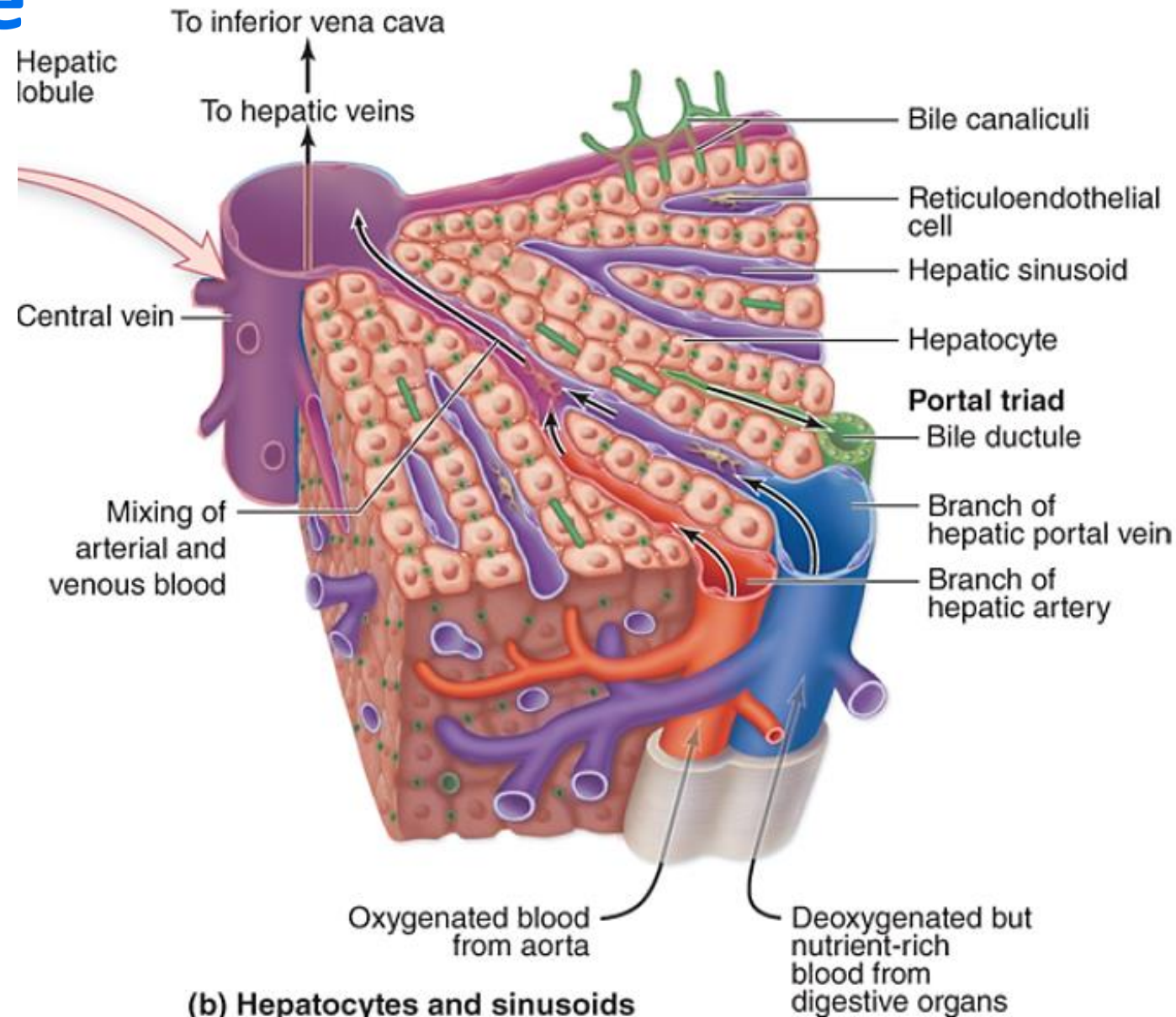
- a) Deoxygenated nutrient-rich supply from the hepatic portal vein.
 - b) Oxygenated blood from the hepatic artery.
- Blood from the portal vein and hepatic artery are mixed in the hepatic sinusoids.



Circulation in the lobule

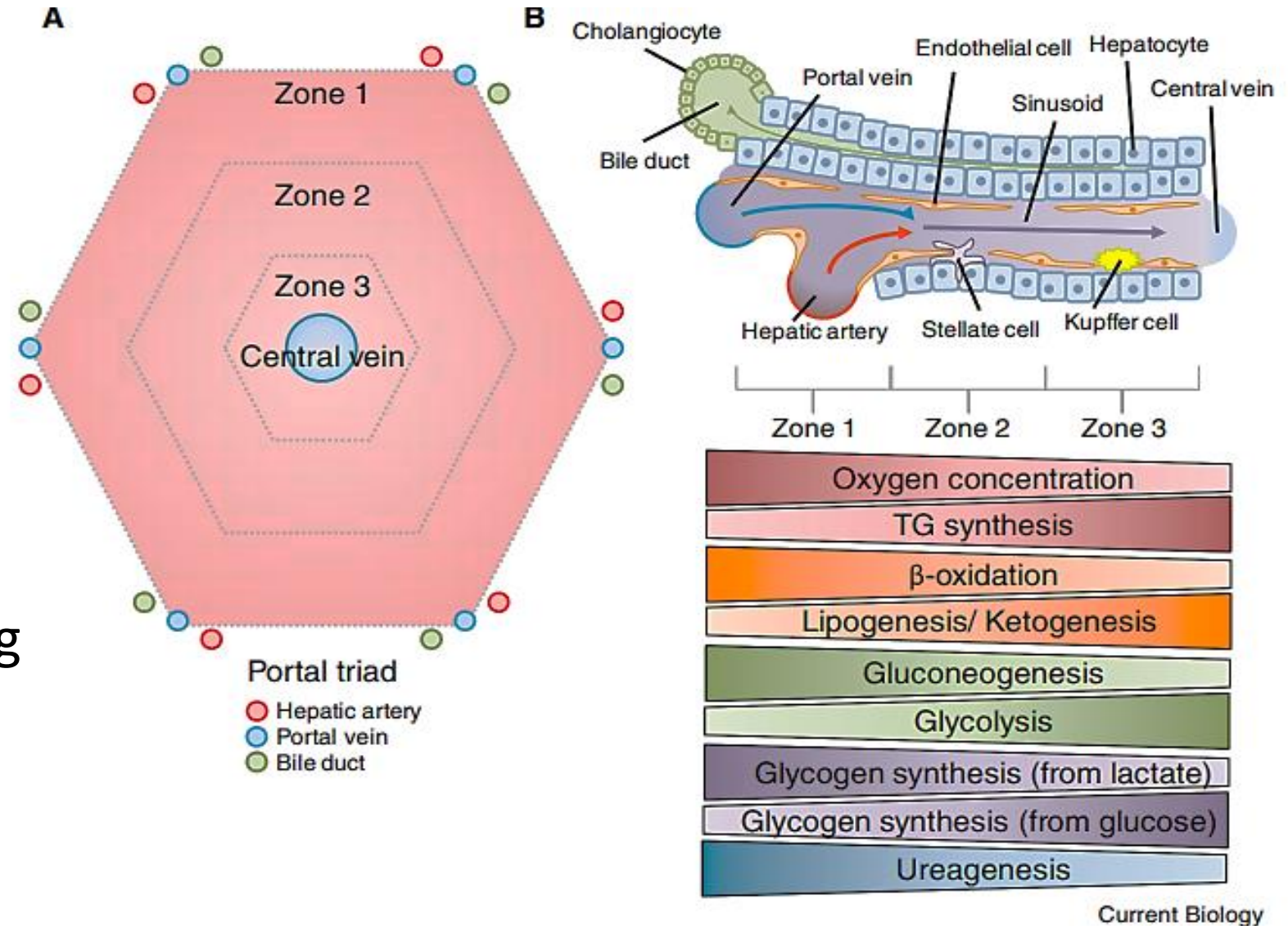
Oxygenated hepatic artery blood mixes with nutrient-rich blood from the portal vein.

Upon mixing, this blood equilibrates and flows across the lobule through a sinusoidal network before draining into branches of the central vein.



Metabolic zonation

As blood moves across the sinusoidal network in the lobules, hepatocytes will utilize oxygen/nutrients and generate waste products. Therefore, hepatocytes will have differential exposure to nutrient and oxygen according to their location, leading to **metabolic zonation** adaptation system.



Autonomic Innervation

- Liver receives efferent fibers of the sympathetic nervous system and parasympathetic nervous system from the splanchnic and vagal nerves, respectively.
- Autonomic innervation aim to control hepatic energy homeostasis, for example:
 - Activation of SNS increase gluconeogenesis and decreased glycogenesis.
 - PNS has the opposite effect (decrease gluconeogenesis and increase glycogenesis).

Regulation of liver mass

Conditional Renewal

Conditional renewal is the capability of hepatocytes to undergo proliferation and renewal in response to injury and cell loss, for example, healthy liver can regenerate itself following partial hepatectomy with the aid of growth factors like, hepatocyte growth factor.

- Liver diseases associated with fibrosis, inflammation, or viral infections impair the regenerative process.
- Liver cells manifest both adaptive hyperplasia and hypertrophy.
 - Adaptive hyperplasia refers to the increase of cell number or proliferation.
 - Hypertrophy refers to the increase in individual cell size due to the increase in cytoplasmic organelles.

Functions of the liver

1- Synthetic function

Production of plasma proteins, complement proteins, coagulation cascade cofactors, and non-essential amino acids.

2- Metabolic function

The liver plays an important role in glucose homeostasis. It can use amino acids as a source of fuel, produce ketone bodies, aids in fatty acid metabolism, synthesis of plasma protein and non-essential amino acids.

3- Storage functions

- Storage Site for Vitamins.

Vitamin A, vitamin D and vitamin B12.

- Storage site for Iron.

Stored in the form of Ferritin. Hepatocytes has apoferritin protein that can reversibly bind to iron and ferritin.

- Storage site for glycogen

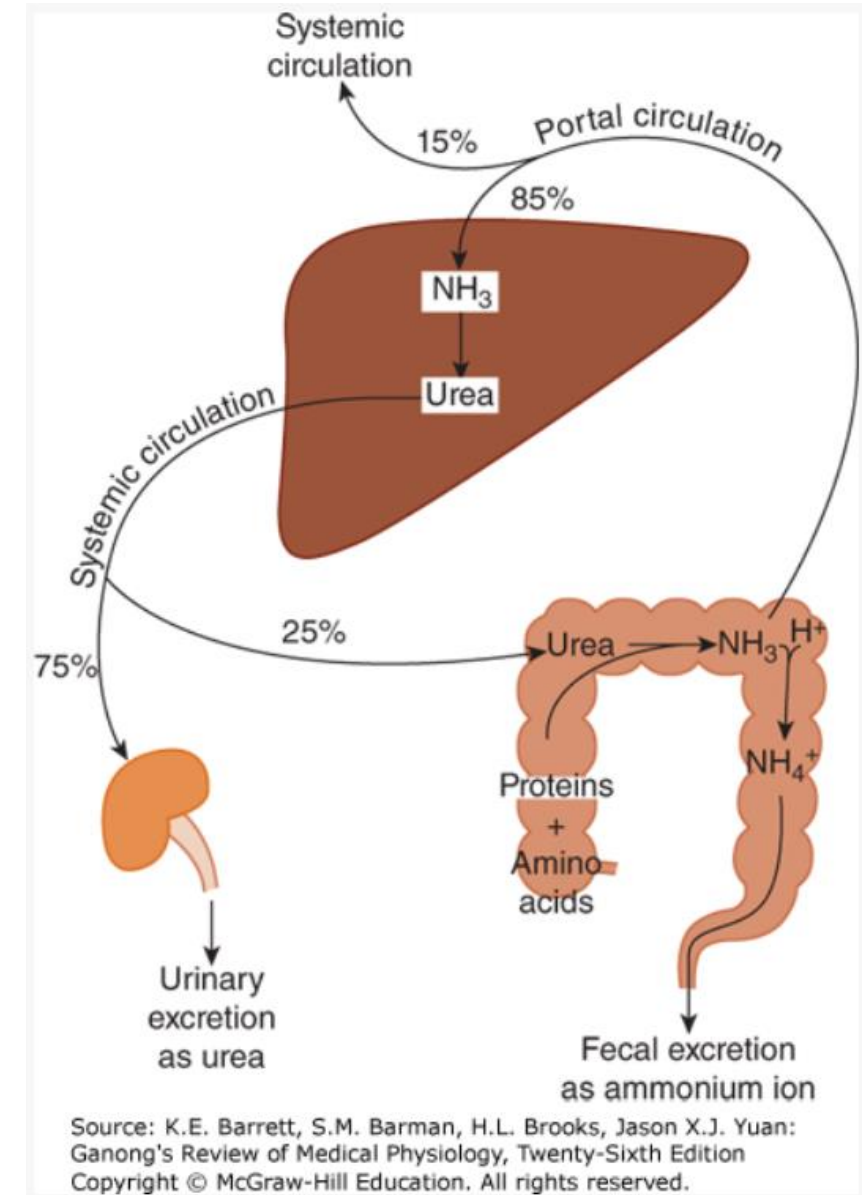
Excess glucose is converted to glycogen and used to maintain glucose homeostasis.

4- Detoxification

Drugs and other compounds (like, ammonia).

The liver detoxifies many substances, including drugs, environmental pollutants, and food additives through phase I and phase II metabolic reaction.

Elevated levels of alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase are indicative of liver injury.



5- Host defense or blood-cleansing function

Through the phagocytic action of Kupffer cells.

Kupffer cells are macrophages that line the hepatic venous sinuses. They efficiently phagocytose bacterium reaching the liver from the portal circulation.

6- Liver clears hormones from the systemic circulation.

This help regulating the regulation of the hormone action.

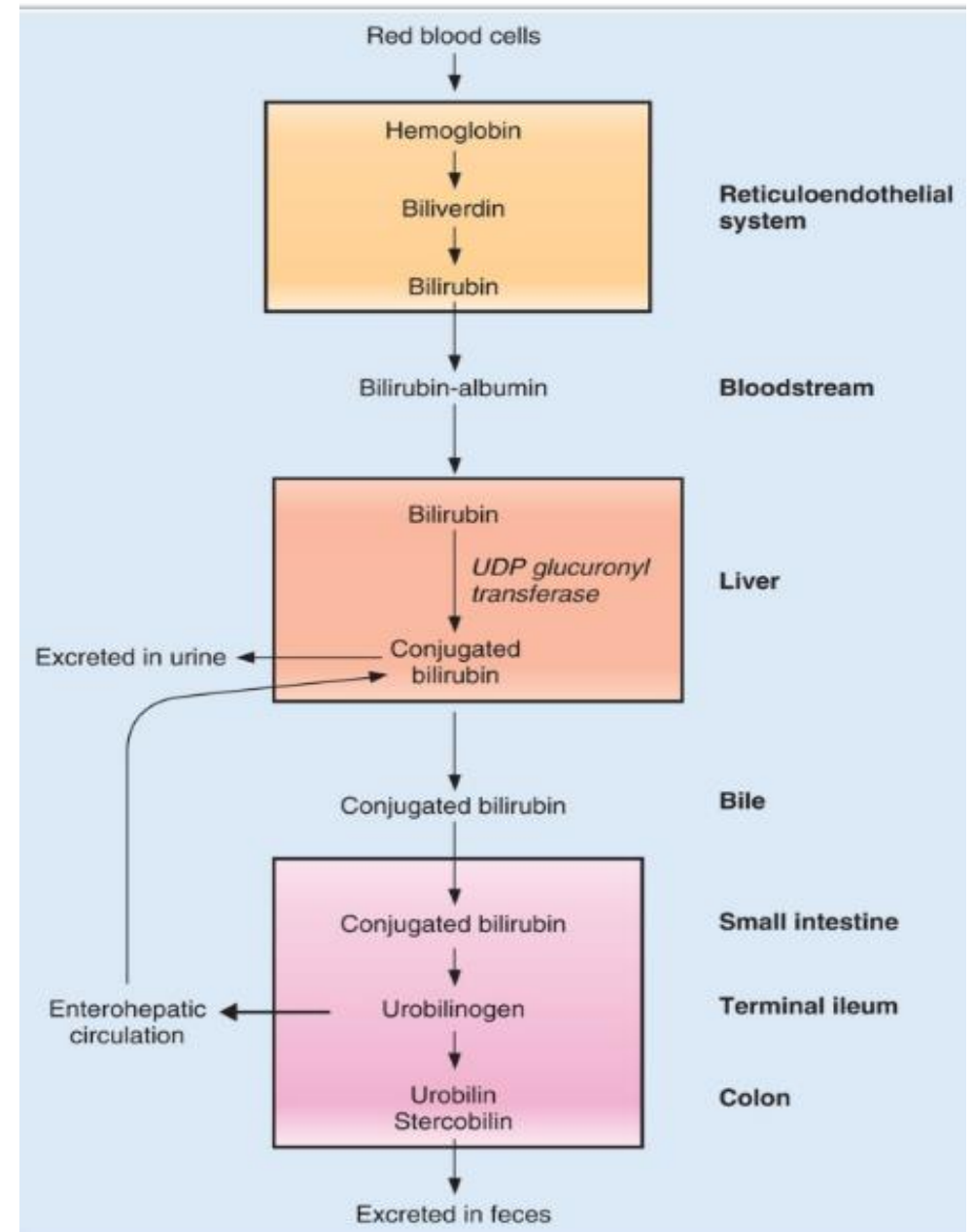
7- Digestive functions “*bile production*”.

Bile formation

Bile contains water, bile salts, phospholipids, cholesterol, and bile pigments (bilirubin).

Around 0.5-1 liter of bile is produced by hepatocytes each day.

In the liver, the synthesized bile drains into the hepatic ducts and is stored in the gallbladder until subsequent release.



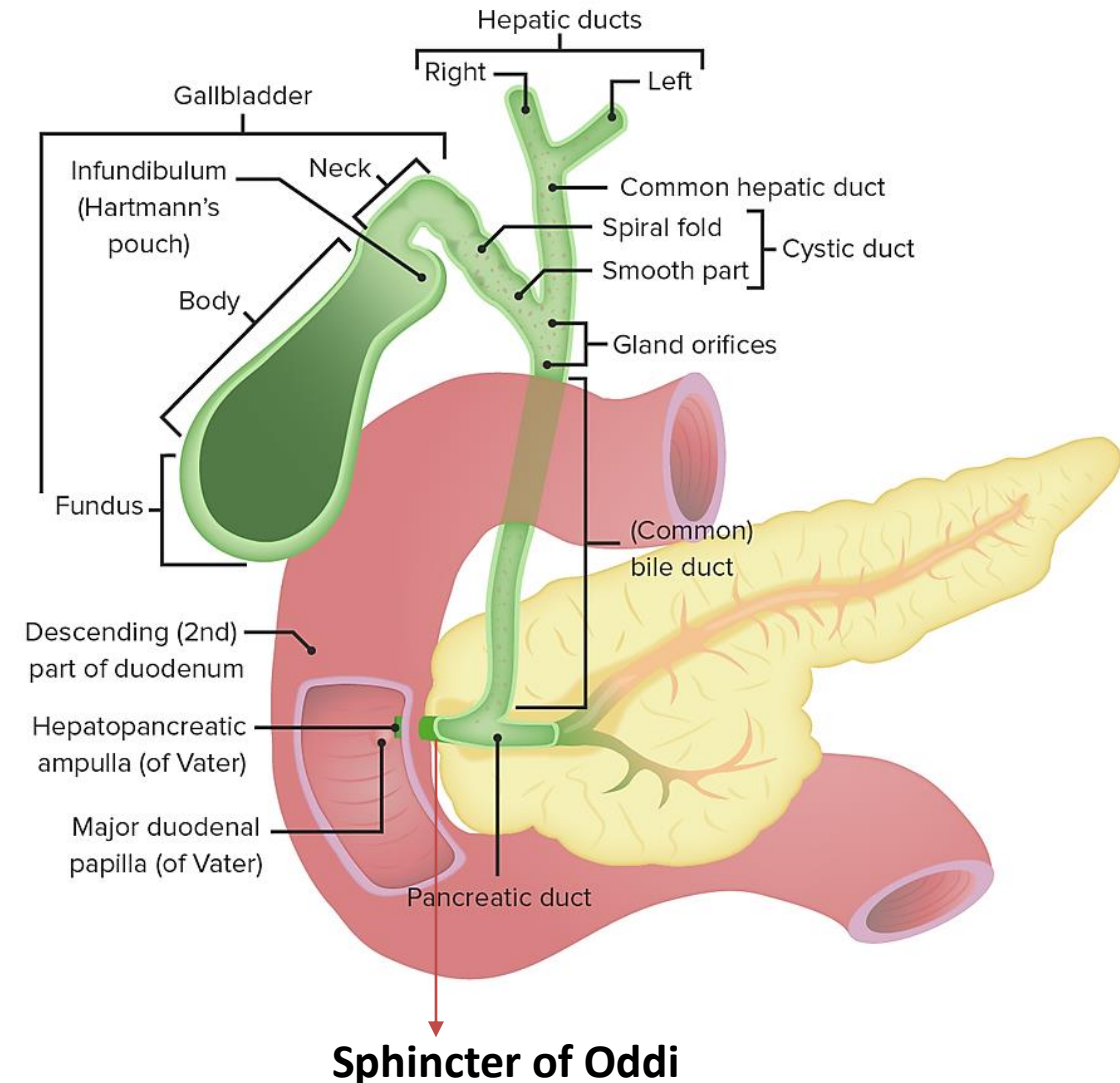
Excess Bilirubin

- Accumulation of free or conjugated bilirubin in the blood will lead to yellowish discoloration of the skin, sclera, and mucous membranes or **jaundice**.
- Diagnosed when the total plasma bilirubin >2 mg/dL.
- **Etiology:**
 - (1) excess production of bilirubin (for example, in hemolytic anemia).
 - (2) decreased uptake of bilirubin into hepatic cells.
 - (3) Disruption of conjugated bilirubin secretion into the bile canaliculi.
 - (4) intrahepatic or extrahepatic bile duct obstruction.

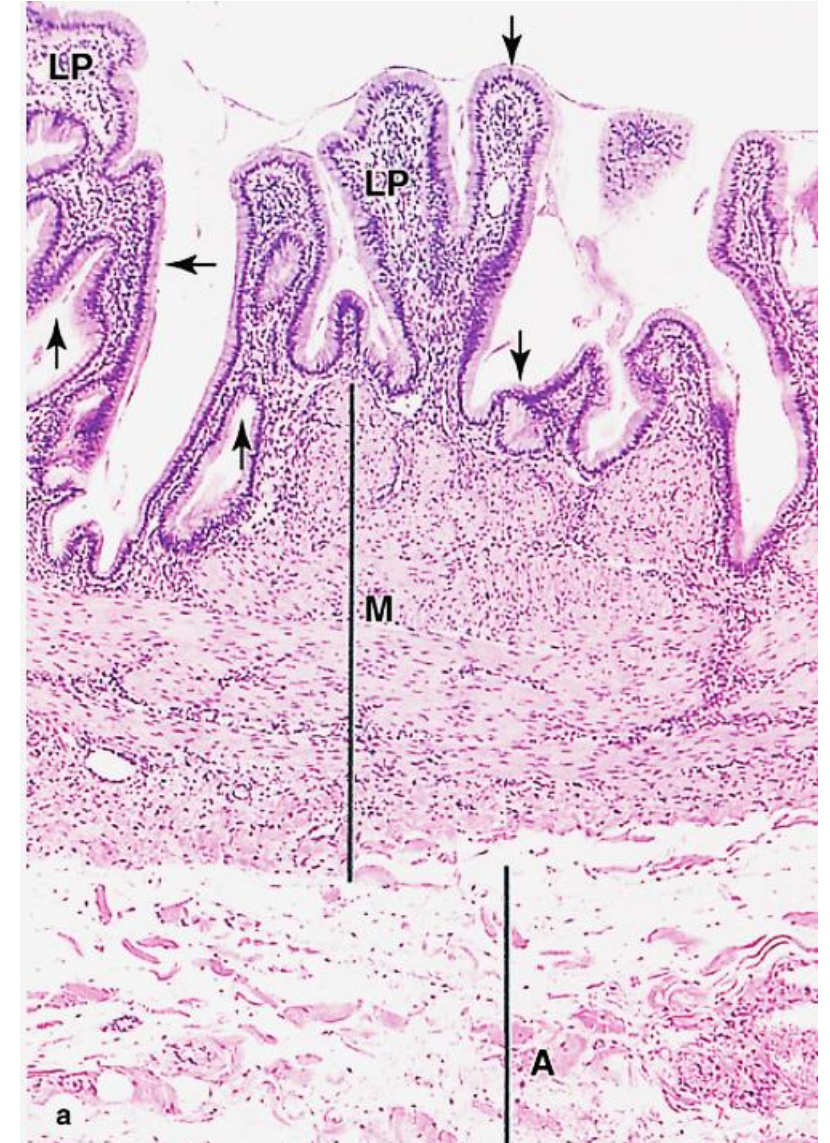


Gallbladder

- The gallbladder is a pear-shaped, sac-like organ.
- It is composed of a fundus, a body and a neck (neck is continuous with the cystic duct).
- Bile flows into the left and right hepatic ducts, merge into the common hepatic duct. Then, it flows to the cystic duct and gallbladder.
- The gall bladder concentrates and stores bile.



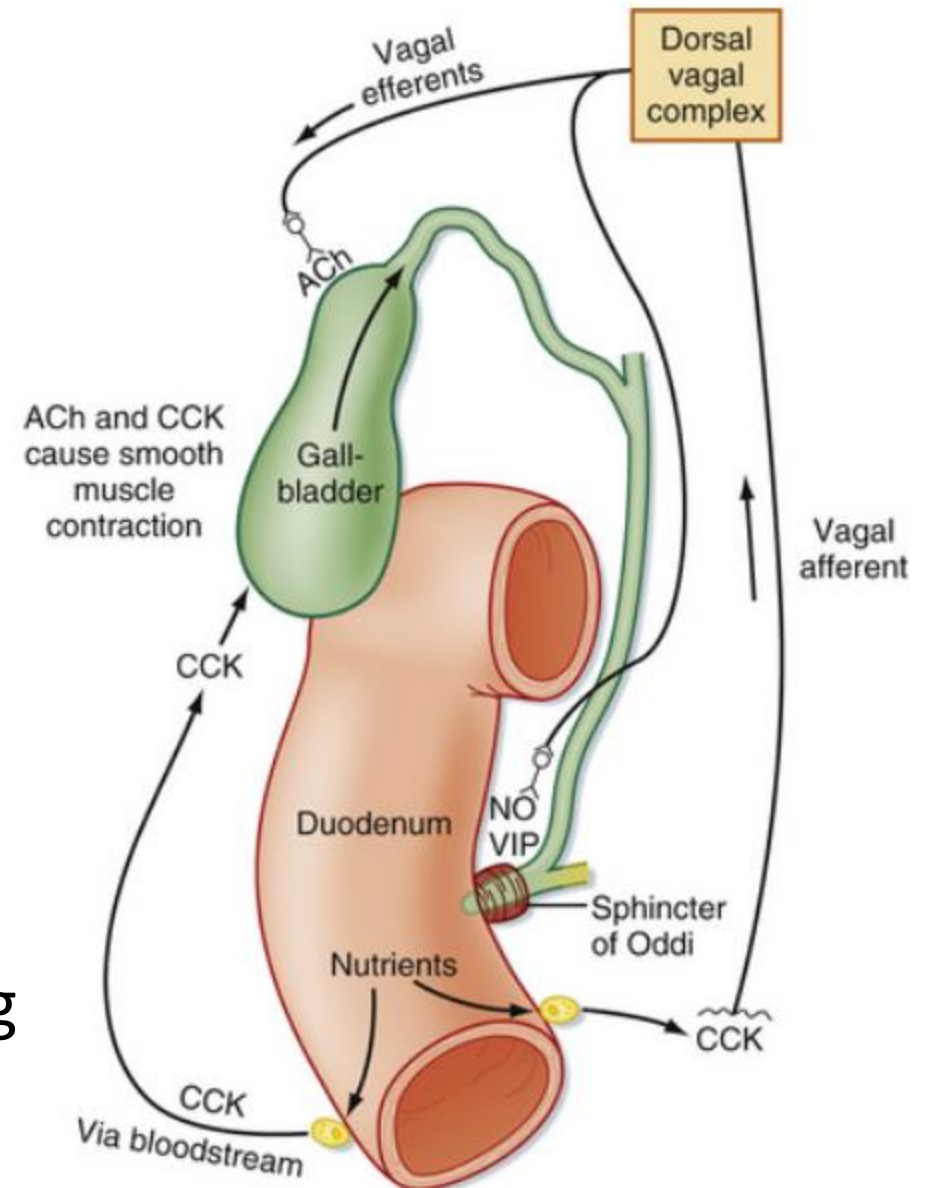
- The mucosa of the gallbladder has many folds (arrows) to increase the surface area. It is lined by columnar epithelia, named, cholangiocytes. They function to uptake the water and concentrate the bile.
- Contraction of the gallbladder muscularis (M) sends bile to the duodenum via the common bile duct.



Regulation of Bile Release

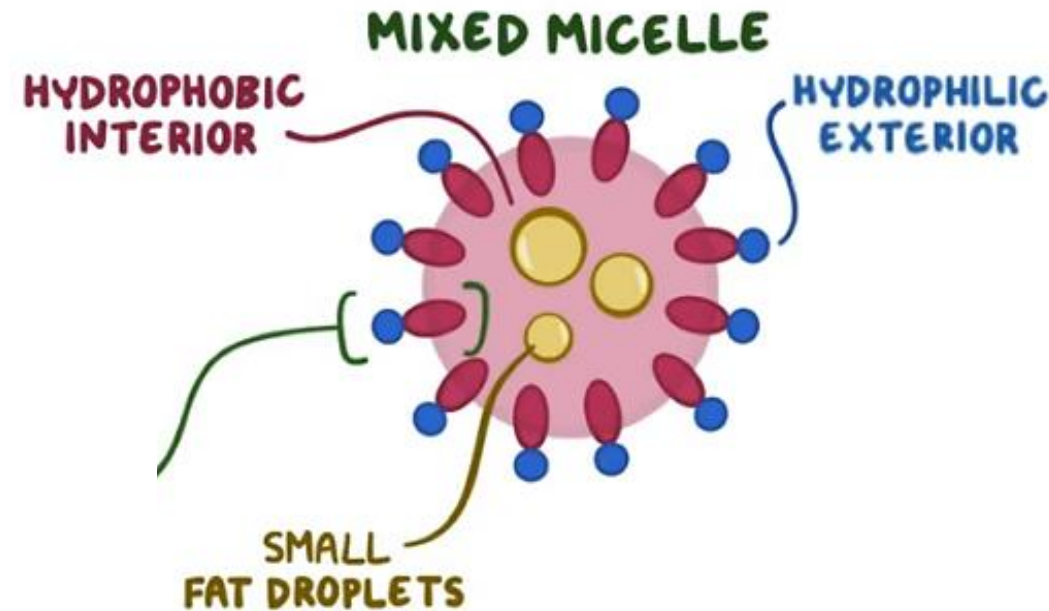
Bile flows into the gallbladder when the sphincter of Oddi is closed (between meals). In the gallbladder, the bile is concentrated by absorption of water.

Following meals, the gallbladder receive hormonal and vagal stimulation via **cholecystikin hormone (CCK)** and Ach, respectively. They cause the gallbladder to contract and sphincter of Oddi to relax, releasing bile into the digestive system.



Functions of bile

- Bile aids in lipid digestion and absorption through micelle formation and emulsification of lipid droplets. This facilitates the absorption of lipids in the small intestine.
- Bile is a route for excretion of waste products from the blood, i.e, bilirubin and excess cholesterol.
- Bile recirculates to the liver via reabsorption from the terminal ileum with Na/bile cotransporter.
- Ileal resection leads to the bile excretion. Consequently, this depletes the bile acid pool possibly leading to steatorrhea.



Liver Function test

- Liver enzymes test: elevated in cases of liver injury.
 - Alanine aminotransferase (ALT; normal 7- 36 U/L) & aspartate aminotransferase (AST; normal 8-48 U/L): increase of serum AST/ALT indicate hepatocyte apoptosis or death. Normal ratio AST/ALT= 1.1:1.
 - Alkaline phosphatase (ALP; normal 40-129 U/L).
- Liver function = synthetic ability. Can be affected hepatic and extrahepatic factors.
 - serum albumin (normal 3.5-5 g/dL)
 - PT or INR for clotting factors.
 - Bilirubin (total: “0.2-1.2 mg/dL; direct “conjugated” <0.3 mg/dl or 30 %; indirect or unconjugated remaining amount). Can indicate pre- vs post-hepatic jaundice.

Resources

- Fish EM, Shumway KR, Burns B. Physiology, Small Bowel. [Updated 2024 Jan 31]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK532263/>

